Renal Physiology

Renal Sodium Handling & ECF Volume Regulation

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Learning Objectives

1. Differentiate the segmental reabsorption of filtered sodium along the nephron, utilizing [TF/P] concentration profiles to evaluate isosmotic fluid reabsorption and preferential solute transport in the proximal tubule.

2. Describe the apical and basolateral transport mechanisms in the early and late proximal tubule, and explain how peritubular capillary Starling forces govern glomerulotubular balance during ECF volume expansion and contraction.

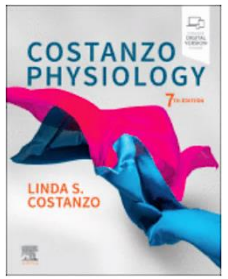
3. Compare the cellular mechanisms of sodium reabsorption in the thick ascending limb, early distal tubule, and collecting duct principal cells, identifying the nephron segment, molecular target, and electrolyte consequences of major diuretic classes.

4. Analyze the systemic regulation of renal sodium excretion, contrasting the actions of the renin-angiotensin-aldosterone system (RAAS), renal sympathetic nerves, and atrial natriuretic peptide (ANP) in defending effective arterial blood volume (EABV).

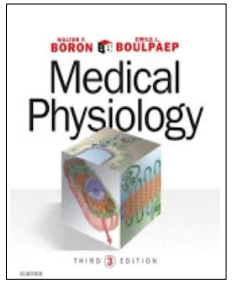
5. Evaluate the integrated physiological adjustments to high- and low-sodium diets, explaining the mechanism and clinical significance of aldosterone escape.

Resources

READ



Chapter 6



Chapter 35

DO

* **TopHat Questions**
- **Guyton and Hall Review Questions (Library)**
- **Costanzo Cases**

TOP HAT

Join Code: 302237

Sodium Balance Governs Extracellular Fluid (ECF) Volume

- **The Master Principle of Renal Sodium Handling:** Sodium balance dictates Extracellular Fluid (ECF) volume, plasma volume, and systemic arterial blood pressure.

- **Sodium Balance Defined:** Daily urinary excretion of Na^+ must exactly match daily dietary Na^+ intake:
  - ***Positive Na^+ Balance:*** **Excretion < Intake** → Na^+ retention → ECF volume expansion, increased arterial pressure, edema.
  - ***Negative Na^+ Balance:*** **Excretion > Intake** → Na^+ loss → ECF volume contraction, decreased arterial pressure, hypovolemia.

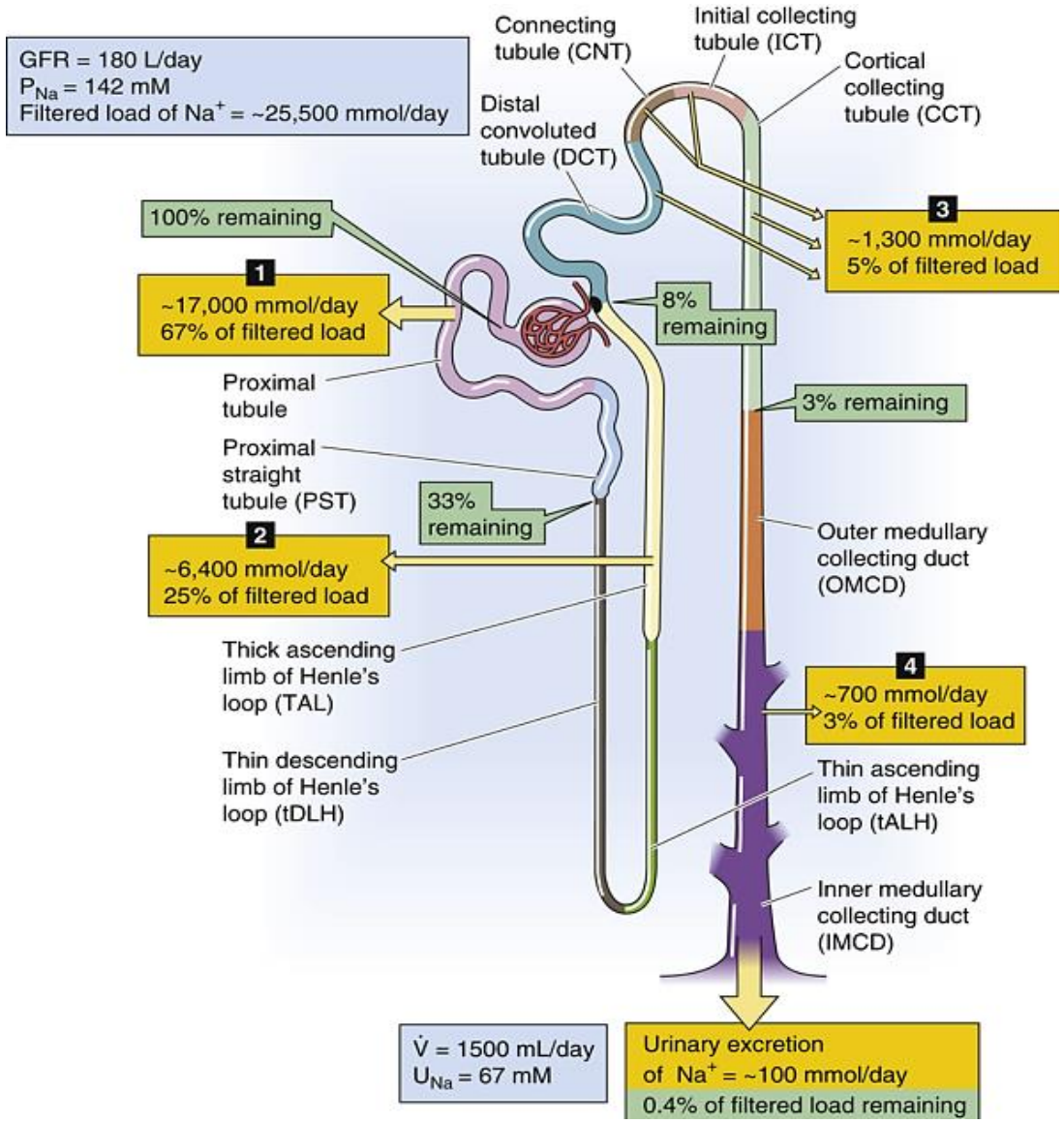
- **Scale of Daily Turnover:**
  - Filtered Load of Na^+ = GFR × P_{Na+} = 180 L/day × 140 mEq/L = 25,200 mEq/day.
  - Normal Daily Intake & Excretion ≈ 150 mEq/day.
  - **Fractional Excretion of Na^+ (FE_{Na+}) < 1%:** The kidneys must reclaim >99% of all filtered sodium every single day.

Segmental Distribution of Sodium Reabsorption

- **Division of Labor Along the Nephron:**
  - Bulk Upstream Reabsorption (PCT + TAL = 92%): relatively less subject to final hormonal fine-tuning.
  - Distal Fine-Tuning (Early DCT + Late DCT/CD = 8%): Regulated segment that establishes homeostatic balance.

- **Nephron Segment Load-Dependency:** Thick ascending limb and distal segments possess load-dependent transport, i.e. if proximal uptake is blocked, downstream segments partially compensate.

- **Final Common Pathway:** Late distal tubule and collecting duct reabsorb only 3%, but this narrow window can determine whether a patient is in hypovolemic shock or hypertensive crisis.



Single Nephron Dynamics: [TF/P] Ratio Concept

- **The Tubular Fluid-to-Plasma [TF/P] Ratio:** Compares solute concentration in tubular fluid at any nephron point to systemic arterial plasma.

- **Interpretive Rules for Solute Handling:**
  - $[TF/P]_x = 1.0$: Solute concentration equals plasma concentration.
  - $[TF/P]_x < 1.0$: Solute has been reabsorbed *faster* than water.
  - $[TF/P_x] > 1.0$: Solute has been reabsorbed *slower* than water, or actively secreted.

[TF/P] Ratio Concept: Continued

Measuring Fractional Water Reabsorption with Inulin:

- Fraction of filtered water remaining $= \frac{1}{[TF/P]_{\text{inulin}}}$
- Fraction of filtered water reabsorbed $= 1 - \frac{1}{[TF/P]_{\text{inulin}}}$
  - If $[TF/P]_{inulin}$ = 2.0 → 50% of water reabsorbed.
  - If $[TF/P]_{inulin}$ = 3.0 (end of PCT) → 67% (two-thirds) of water reabsorbed.

Fraction of Filtered Solute Remaining:

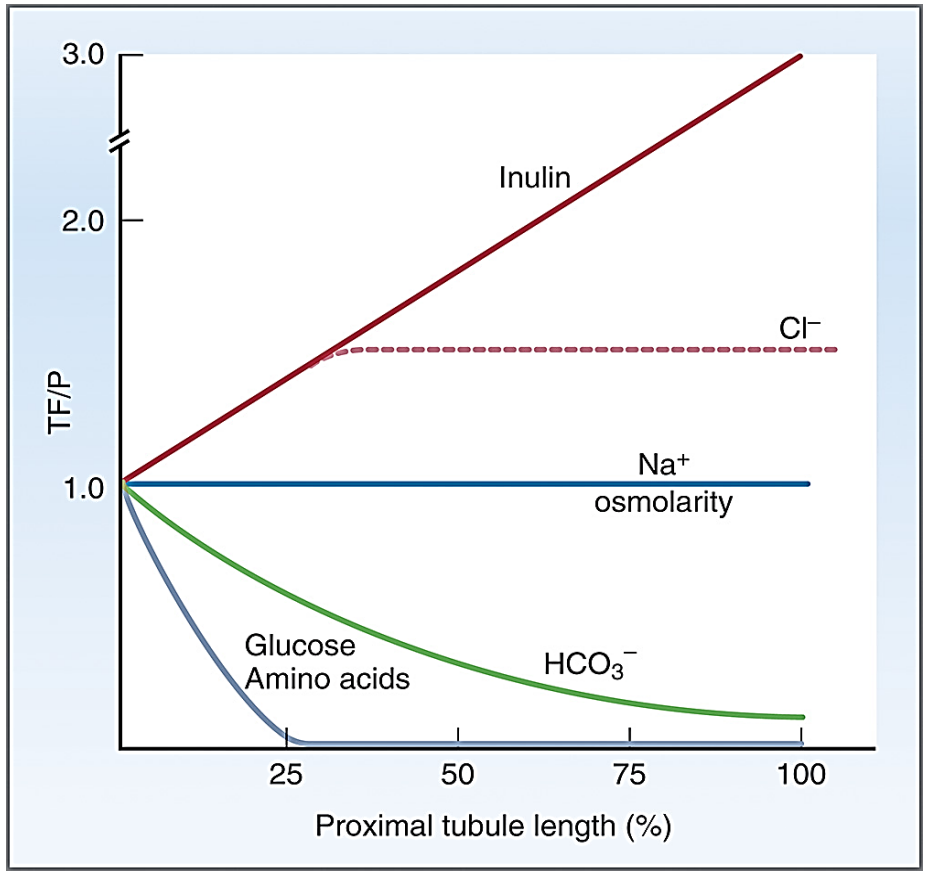
- Fraction of Solute Remaining $= \frac{[\text{TF/P}]_x}{[\text{TF/P}]_{\text{inulin}}}$
  - At end of PCT, $[TF/P]_{Na+}$ = 1.0, but $[TF/P]_{Na+}$ / $[TF/P]_{inulin}$ = 1.0 / 3.0 = 0.33 (33% remains; 67% reabsorbed).

[TF/P] Ratio Concept: Examples

- $[TF/P]_{\text{inulin}} = \frac{[\text{Inulin}]_{\text{tubular fluid}}}{[\text{Inulin}]_{\text{plasma}}} : [TF/P]_{\text{inulin}} = \frac{2}{1} = 2$
  - This means tubular-fluid inulin concentration is twice the plasma concentration.
- Fraction of filtered water remaining $= \frac{1}{[TF/P]_{\text{inulin}}} = \frac{1}{2} = 0.50$
  - So 50% of the filtered water has been reabsorbed.
- Fraction of filtered solute x remaining $= \frac{[TF/P]_x}{[TF/P]_{\text{inulin}}} = \frac{0.5}{2} = 0.25$
  - So 25% of the filtered solute remains in the tubular fluid.
- Fraction of filtered solute x reabsorbed $= 1 - \frac{[TF/P]_x}{[TF/P]_{\text{inulin}}} = 1 - \frac{0.5}{2} = 0.75$
  - So 75% of the filtered solute has been reabsorbed.

[TF/P] Profiles Along the Proximal Tubule

- **Preferential Early Reabsorption:**
  - In the PCT, Na^+ cotransport rapidly strips the lumen of 100% of filtered glucose and amino acids.
  - Na^+/H^+ exchange preferentially reclaims 85% of filtered HCO_3^-.
- **The Reciprocal Behavior of Chloride:**
  - Because HCO_3^- is cleared early while Cl^- is left behind, water reabsorption concentrates luminal Cl^-.
  - $[TF/P]_{Cl^-}$ rises to ~1.3–1.4, creating the chemical gradient that powers late PCT NaCl reabsorption.
- **Takeaway:** The proximal tubule maintains isosmotic volume reduction while sequentially altering luminal solute composition.



Interim Summary 1

- **Volume vs. Concentration:** Total body Na^+ *content* governs extracellular fluid (ECF) volume and blood pressure, whereas body *water* governs plasma [Na^+].

- **Nephron Workload Partitioning:**
  - *Proximal Convoluted Tubule (PCT):* Reabsorbs **~67%** of filtered Na^+ isosmotically with water.
  - *Thick Ascending Limb (TAL):* Reabsorbs **~25%** of filtered Na^+ without water (diluting segment).
  - *Distal Convoluted Tubule (DCT):* Reabsorbs **~5%** of filtered Na^+.
  - *Late DCT & Collecting Duct (CD):* Reabsorbs the final **~3%** via hormonal fine-tuning (urinary excretion < 1%).

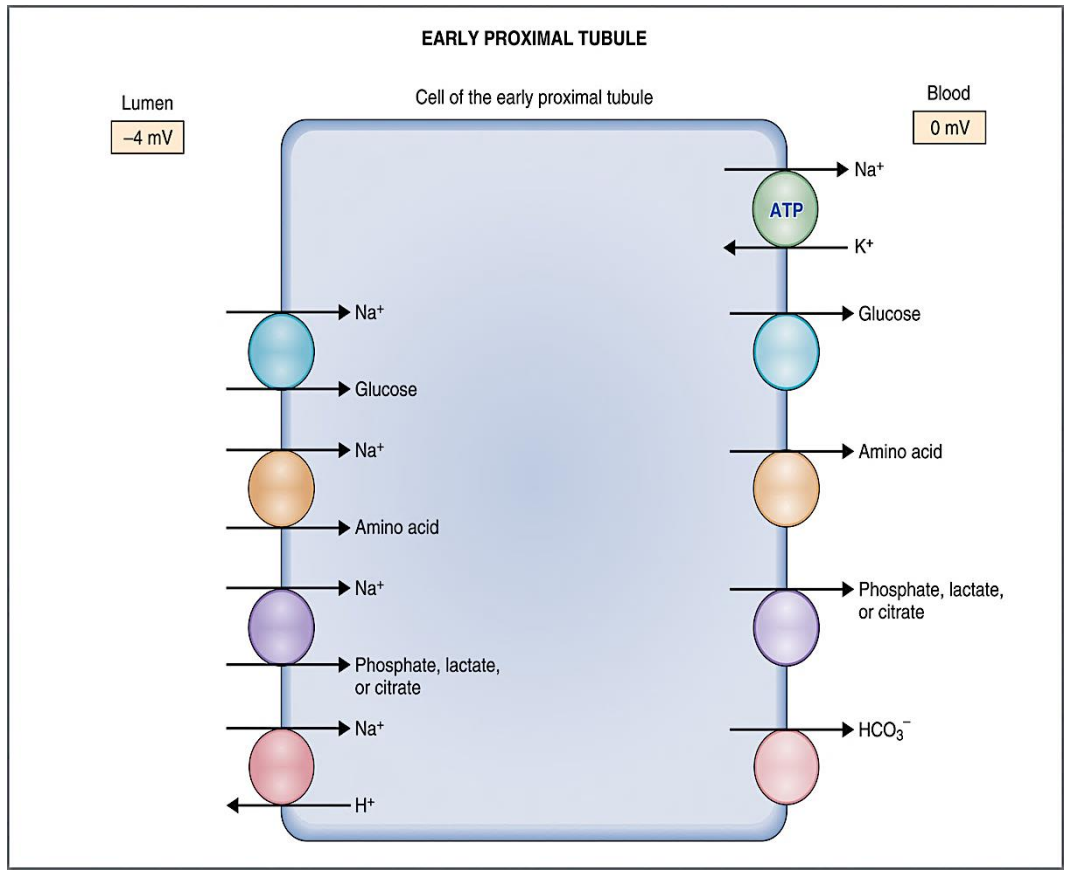
- **[TF/P] Solute Interpretation:**
  - **$[TF/P]_{Na^+}$ = 1.0 throughout the PCT:** Na^+ is reabsorbed in exact equimolar proportion to water (isosmotic).
  - **$[TF/P]_{glucose}$ & $[TF/P]_{AA}$ drop rapidly to 0:** Actively cleared in the first 25% of the PCT.
  - **$[TF/P]_{HCO_3^-}$ drops below 1.0:** Preferentially reabsorbed in early PCT via Na^+/H^+ exchange.
  - **$[TF/P]_{Cl^-}$ rises above 1.0 (~1.3–1.4):** Cl^- is left behind as HCO_3^- and water leave early, creating the chemical gradient that drives late PCT reabsorption.
  - **$[TF/P]_{inulin}$ rises from 1.0 to 3.0:** Inulin is not reabsorbed; a 3-fold concentration rise confirms that two-thirds (67%) of filtered water was reabsorbed.

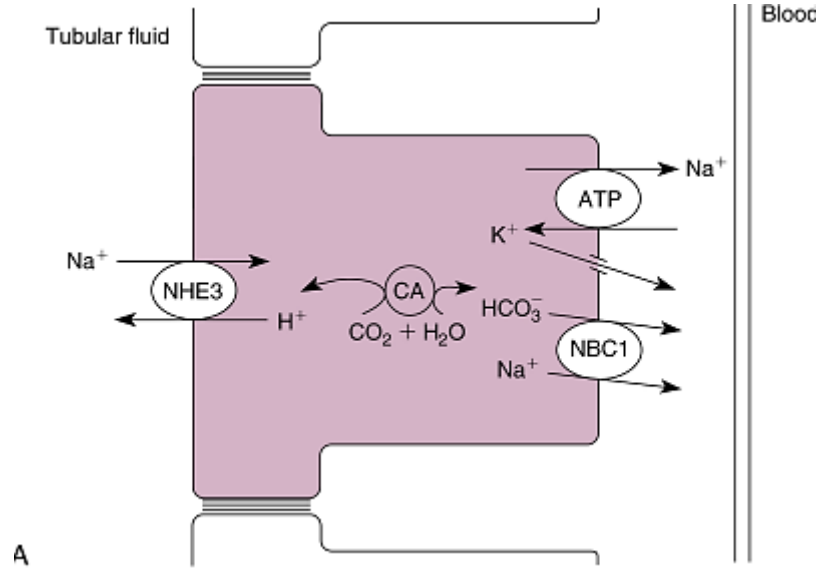
Early PCT: Solute Cotransport & NHE3

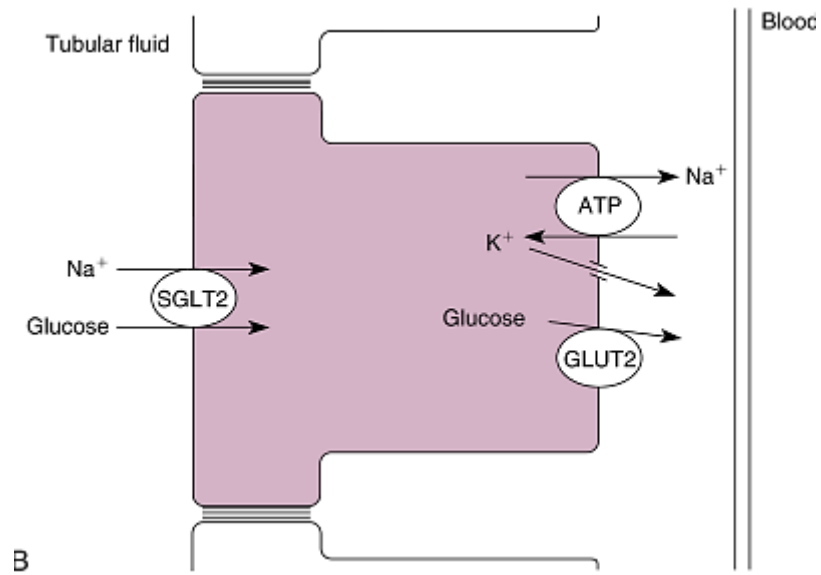
- **The Universal Driving Engine:** Basolateral Na^+/K^+-ATPase hydrolyzes ATP to maintain low intracellular [Na^+] (~10–15 mM) and negative voltage (-70 mV), generating the chemical gradient that powers all apical entry.

- **Secondary Active Cotransport:**
  - Downhill Na^+ entry pulls glucose (SGLT2), amino acids, and phosphate uphill into the cell.
  - Solutes exit the basolateral membrane via passive facilitated diffusion.

- **Sodium-Hydrogen Exchange (NHE3):**
  - Extrudes H^+ to titrate luminal HCO_3^- to CO_2 and H_2O via brush border carbonic anhydrase.
  - Intracellular reassembly yields Na^+ and HCO_3^- reabsorption into peritubular blood.
  - **Pharmacology:** Carbonic anhydrase inhibitors (Acetazolamide) block this cascade, causing mild natriuresis, bicarbonate wasting, and metabolic acidosis.


EARLY PROXIMAL TUBULE
Lumen
–4 mV
Cell of the early proximal tubule
Blood
0 mV
Na+
Glucose
Na+
Amino acid
Na+
Phosphate, lactate, or citrate
Na+
H+
ATP
Na+
K+
Glucose
Amino acid
Phosphate, lactate, or citrate
HCO3−


Tubular fluid
Blood
Na+
ATP
K+
Na+
NHE3
H+
CA
CO2 + H2O
HCO3−
NBC1
Na+
A

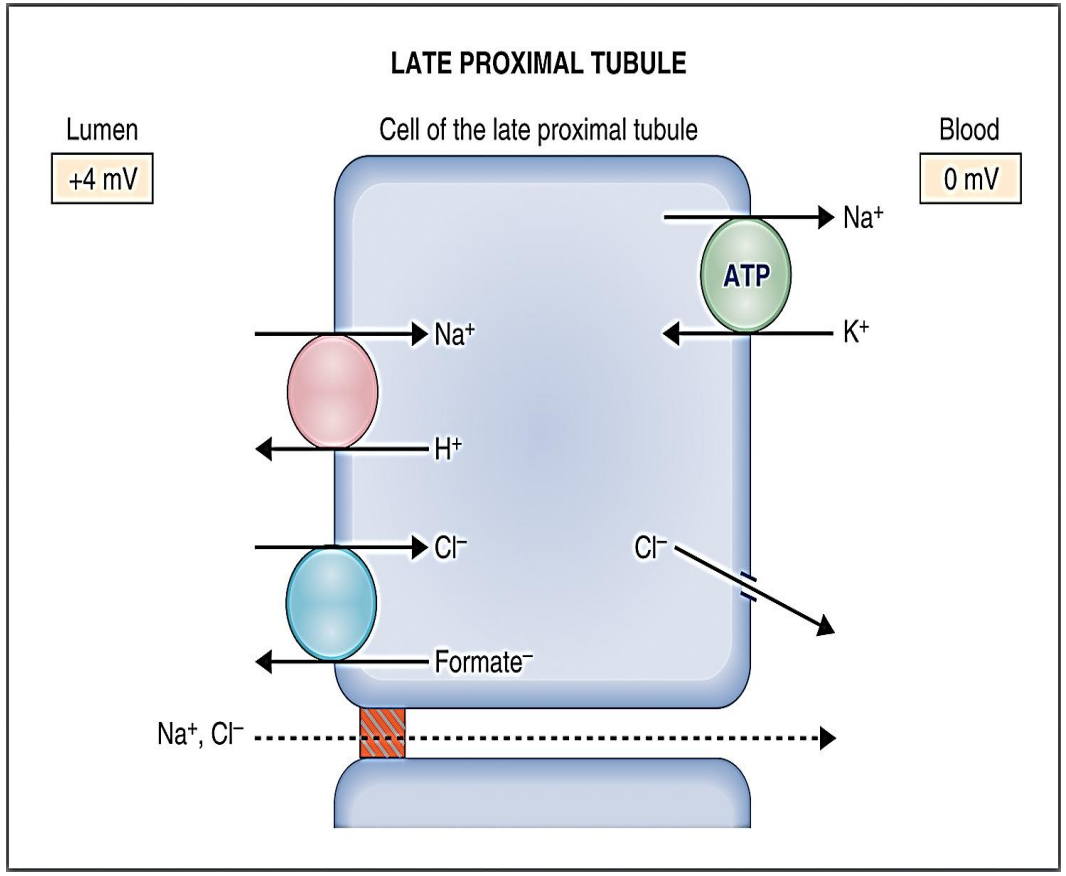

Tubular fluid
Blood
Na+
ATP
K+
Na+
SGLT2
Glucose
Glucose
GLUT2
B

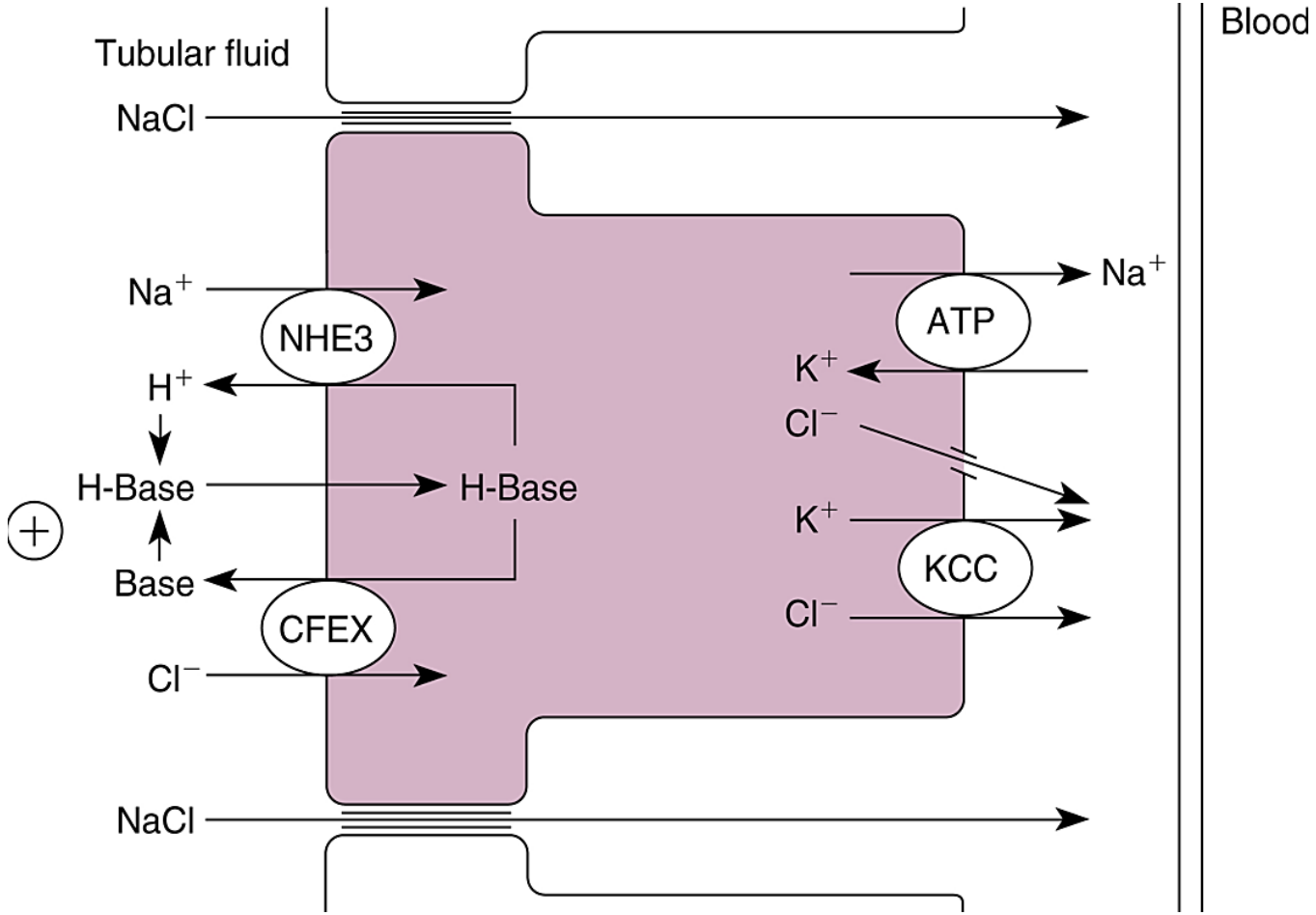
Late PCT: Chloride-Driven Reabsorption

- **Shift in Luminal Fluid Composition:** By the mid-PCT, glucose and amino acids are gone, and luminal [Cl^-] is elevated (~140 mEq/L vs. ~105 mEq/L in plasma).

- **Cellular Reabsorption Route:**
  - NHE3 extrudes H^+ while the Cl^-/formate$^-$ exchanger brings in Cl^-.
  - Intracellular formate recycles across the apical membrane, resulting in net electroneutral cellular uptake of NaCl.

- **Paracellular Reabsorption Route (The Leaky Shunt):**
  - The high luminal [Cl^-] drives passive Cl^- diffusion across leaky tight junctions into the interstitium.
  - Negative chloride ions leaving the lumen create a **lumen-positive transepithelial potential (+4 mV)**.
  - This positive lumen electrostatically repels Na^+, driving passive paracellular Na^+ reabsorption.


LATE PROXIMAL TUBULE
Lumen
+4 mV
Cell of the late proximal tubule
Blood
0 mV
Na+
ATP
K+
Na+
H+
Cl−
Cl−
Formate−
Na+, Cl−


Tubular fluid
Blood
NaCl
Na+
NHE3
H+
H-Base
H-Base
Base
CFEX
Cl−
NaCl
Na+
ATP
K+
Cl−
K+
KCC
Cl−

Glomerulotubular Balance & Starling Forces

- **Glomerulotubular (GT) Balance Defined:**
  The intrinsic tendency of the proximal tubule to reabsorb a relatively constant fraction (~65–67%) of the filtered Na^+ and water load as filtered load changes.

- GT balance refers to the fact that Na^+ and water reabsorption increases in proportion to the increase in GFR and filtered amount of Na^+.
  - Thus a constant fraction of filtered Na^+ and water is reabsorbed from the proximal tubule despite variations in GFR.
  - Net result of GT balance is to reduce the impact of GFR changes on the amount of Na^+ and water excreted in the urine when the extracellular fluid volume is normal.

- **Load-Dependent Proximal Reabsorption:**
  ↑ Filtered Na^+ and water load → ↑ absolute proximal reabsorption
  ↓ Filtered Na^+ and water load → ↓ absolute proximal reabsorption

Glomerulotubular Balance & Starling Forces

- **Peritubular Starling Forces Support Reabsorption:**
  Fluid reabsorbed from proximal tubule enters the interstitium and is taken up by peritubular capillaries.

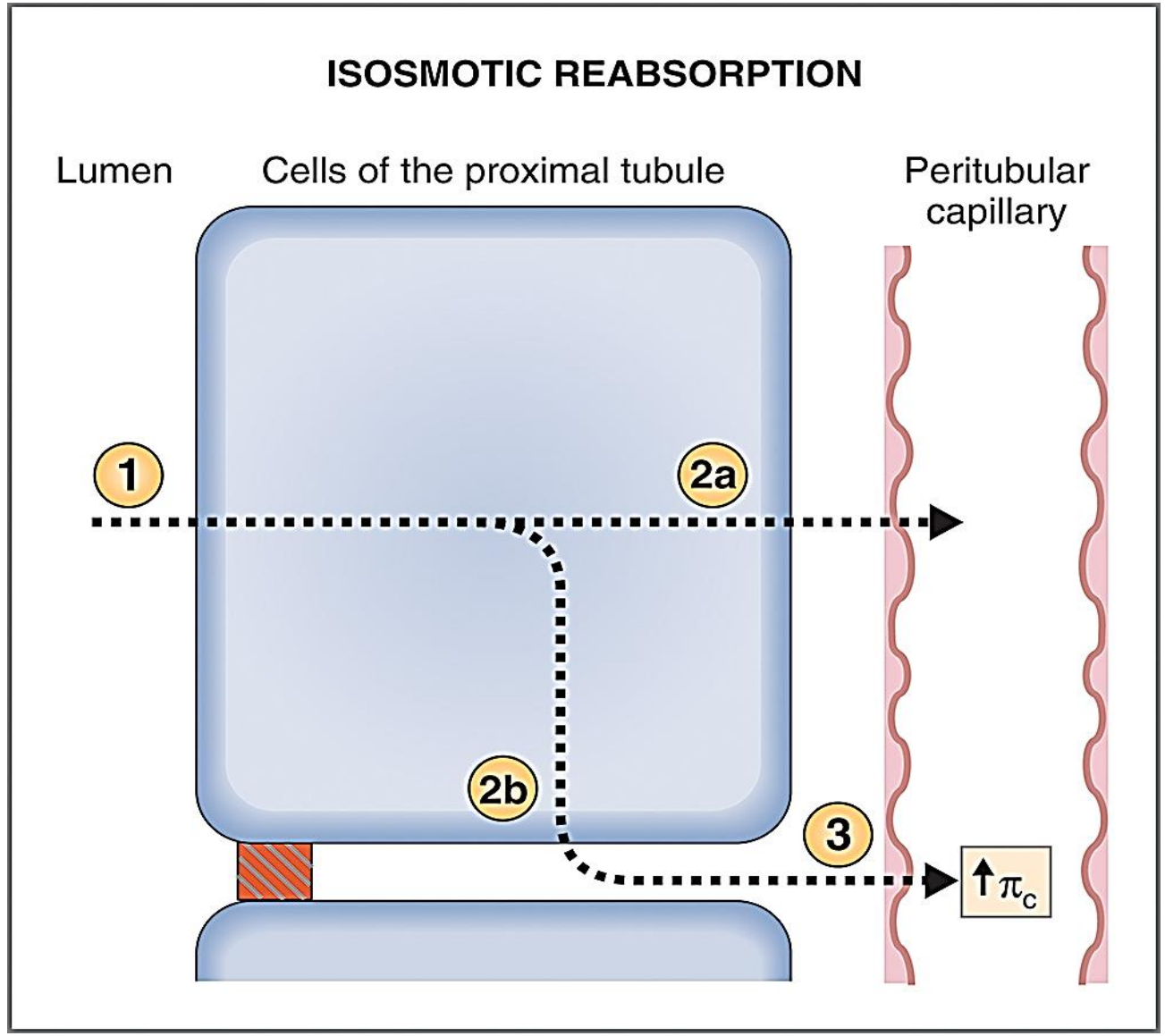
Peritubular uptake is favored by:

- **High peritubular capillary oncotic pressure (↑ π_c)**
- **Low peritubular capillary hydrostatic pressure (↓ P_c)**

Role of Filtration Fraction:
FF = GFR / RPF

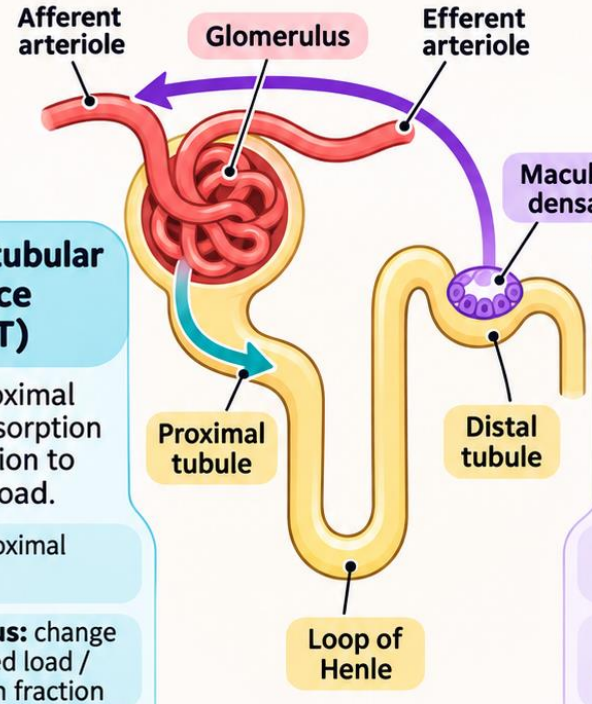
- When **FF increases**, more plasma water is filtered relative to plasma flow while proteins remain in the efferent blood:
  - **↑ FF → ↑ efferent/peritubular protein concentration → ↑ π_c → ↑ proximal fluid reabsorption**

- **Physiological Protection:**
  Glomerulotubular balance limits large changes in distal Na^+ and water delivery when filtered load changes.


ISOSMOTIC REABSORPTION
Lumen
Cells of the proximal tubule
Peritubular capillary
1
2a
2b
3
↑πc

Glomerulotubular Balance vs Tubuloglomerular Feedback

Same nephron, different direction, different function



Glomerulotubular balance (G → T)

Adjusts proximal tubular reabsorption in proportion to filtered load.

- **Site:** proximal tubule
- **Stimulus:** change in filtered load / filtration fraction
- **Purpose:** prevents excessive salt and water loss

Tubuloglomerular feedback (T → G)

Adjusts GFR based on NaCl delivery to the macula densa.

- **Site:** macula densa → afferent arteriole
- **Stimulus:** distal NaCl delivery
- **Purpose:** autoregulates GFR

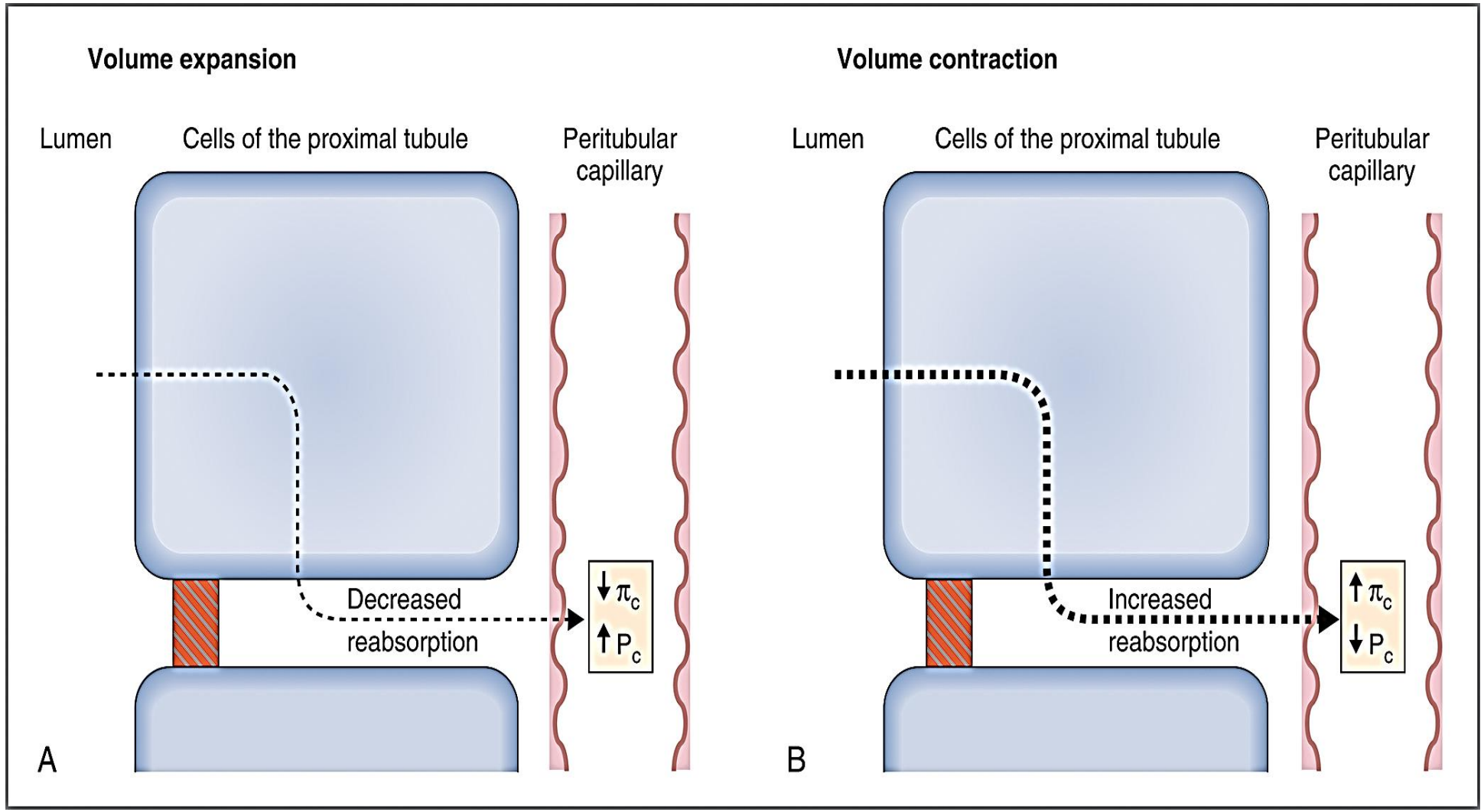
	Glomerulotubular Balance (G → T)	Tubuloglomerular Feedback (T → G)
Input signal	Filtered load / FF	Macula densa NaCl
Target	Proximal tubule	Afferent arteriole / GFR
Output	Reabsorption changes	GFR changes

ECF Volume Shifts: Expansion vs. Contraction

- **ECF Volume Expansion** modifies the set point/fraction of proximal reabsorption:
  - Isotonic saline infusion dilutes systemic plasma proteins (↓ π_c) and raises vascular hydrostatic pressure (↑ P_c).
  - Opposes bulk peritubular capillary uptake; isosmotic fluid leaks backwards across leaky tight junctions into the lumen.
  - **Result:** Decreased fractional Na^+ reabsorption (<67%), promoting pressure natriuresis.

- **ECF Volume Contraction Maximizes Reabsorption:**
  - Fluid loss (diarrhea, vomiting, sweating) hemoconcentrates plasma proteins (↑ π_c) and lowers capillary pressure (↓ P_c).
  - Enhances Starling reabsorptive driving force; fractional reabsorption climbs (>67%).

- **Endocrine Synergism & Contraction Alkalosis:**
  - Low ECF volume triggers Angiotensin II, which directly stimulates apical NHE3.
  - Stimulated NHE3 co-reabsorbs Na^+, water, and HCO_3^-, generating **contraction alkalosis**.


Volume expansion
Lumen
Cells of the proximal tubule
Peritubular capillary
Decreased reabsorption
↓ π_c
↑ P_c
A
Volume contraction
Lumen
Cells of the proximal tubule
Peritubular capillary
Increased reabsorption
↑ π_c
↓ P_c
B

Interim Summary 2

- **Early PCT Transport:** Powered by the basolateral Na^+/K^+-ATPase. Apical Na^+ entry drives secondary active cotransport of glucose (SGLT2), amino acids, and phosphate, alongside countertransport of H^+ via NHE3 for HCO_3^- reclamation (-4 mV lumen potential).

- **Late PCT Transport:** High luminal [Cl^-] drives electroneutral transcellular NaCl uptake (NHE3 coupled to Cl^-/formate$^-$ exchange) and passive paracellular Cl^- diffusion. Cl^- leaving the lumen generates a **+4 mV lumen-positive potential**, electrostatically repelling Na^+ through leaky tight junctions.

- **Glomerulotubular (GT) Balance:** The proximal tubule automatically maintains constant fractional reabsorption (~67%) when GFR fluctuates.
  - ↑ GFR → ↑ Filtration Fraction (FF = GFR / RPF) → ↑ peritubular capillary oncotic pressure (↑ π_c) and ↓ hydrostatic pressure (↓ P_c) → ↑ peritubular fluid uptake.

- **ECF Volume Shifts Override GT Balance:**
  - *ECF Volume Expansion:* Dilutes peritubular proteins (↓ π_c) and raises capillary pressure (↑ P_c) → causes tight-junction back-leak, dropping fractional Na^+ reabsorption below 67% (pressure natriuresis).
  - *ECF Volume Contraction:* Hemoconcentrates proteins (↑ π_c) and activates Angiotensin II (stimulating NHE3) → maximizes fluid uptake (> 67%) and produces **contraction alkalosis**.

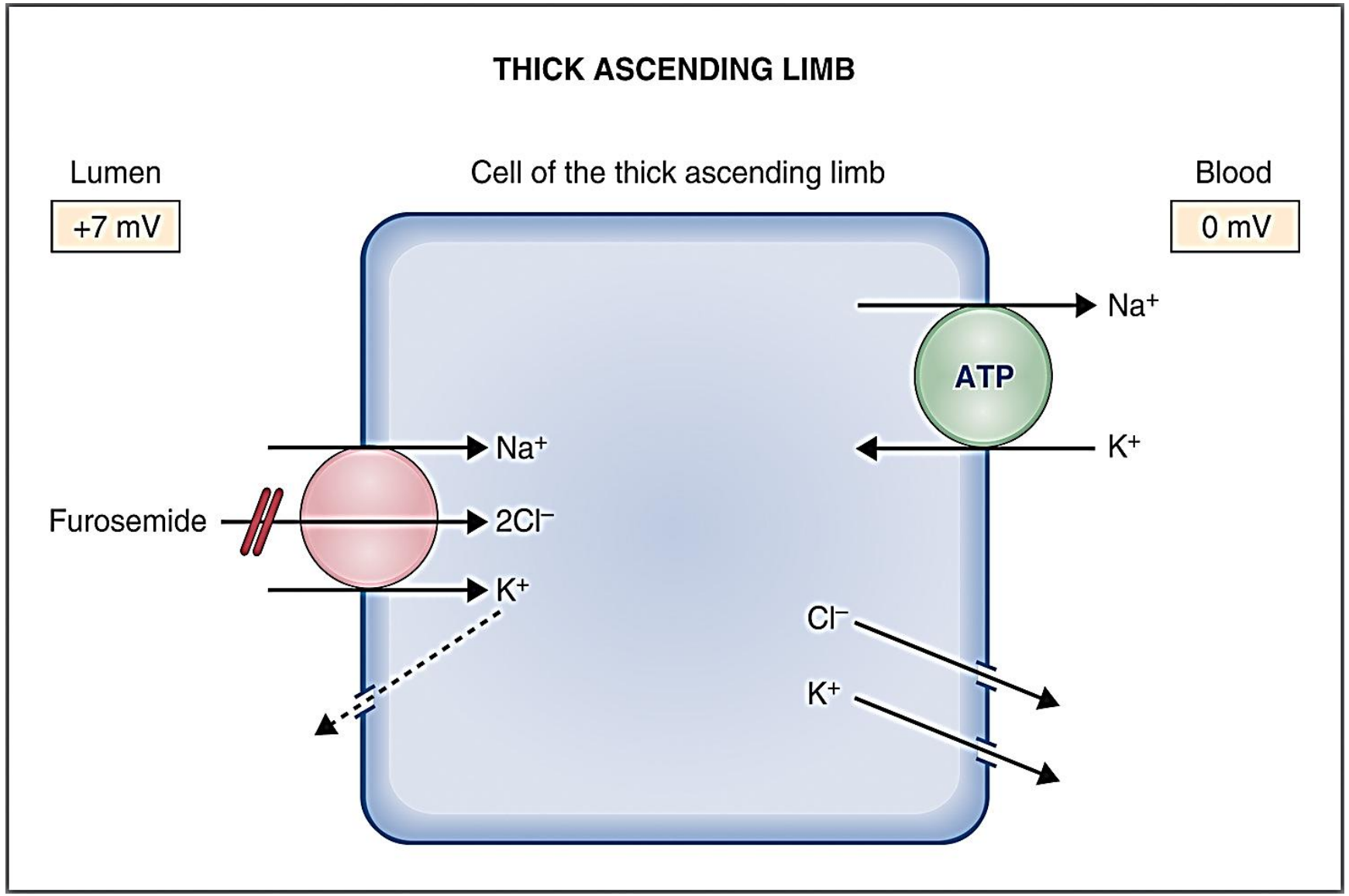
Thick Ascending Limb (TAL): NKCC2 Engine

- **NKCC2 cotransporter:** Reabsorbs 25% of filtered Na^+ via the electroneutral apical NKCC2 cotransporter.

- **Watertight Epithelium ("The Diluting Segment"):** Completely impermeable to water; active solute extrusion dilutes luminal fluid to ~100 mOsm/L while salting the medullary interstitium.

- **Generation of the Lumen-Positive Potential (+7 mV):**
  - Intracellular K^+ brought in by NKCC2 recycles back into the lumen via apical ROMK channels.
  - This net back-diffusion of positive charge creates a **+7 mV lumen-positive potential**.
  - **Clinical High-Yield:** The +7 mV charge electrostatically repels divalent cations, driving **paracellular Ca^{2+} and Mg^{2+} reabsorption**.

- **Loop Diuretics (Furosemide, Bumetanide, Torsemide):**
  - Compete for the luminal Cl^--binding site on NKCC2.
  - Abolishes the +7 mV potential → causes profound **wasting of Na^+, Cl^-, K^+, Ca^{2+}, and Mg^{2+}**.
  - May be administered for acute pulmonary edema and acute hypercalcemia.


THICK ASCENDING LIMB
Lumen
+7 mV
Cell of the thick ascending limb
Blood
0 mV
Na+
ATP
K+
Na+
Furosemide
2Cl−
K+
Cl−
K+

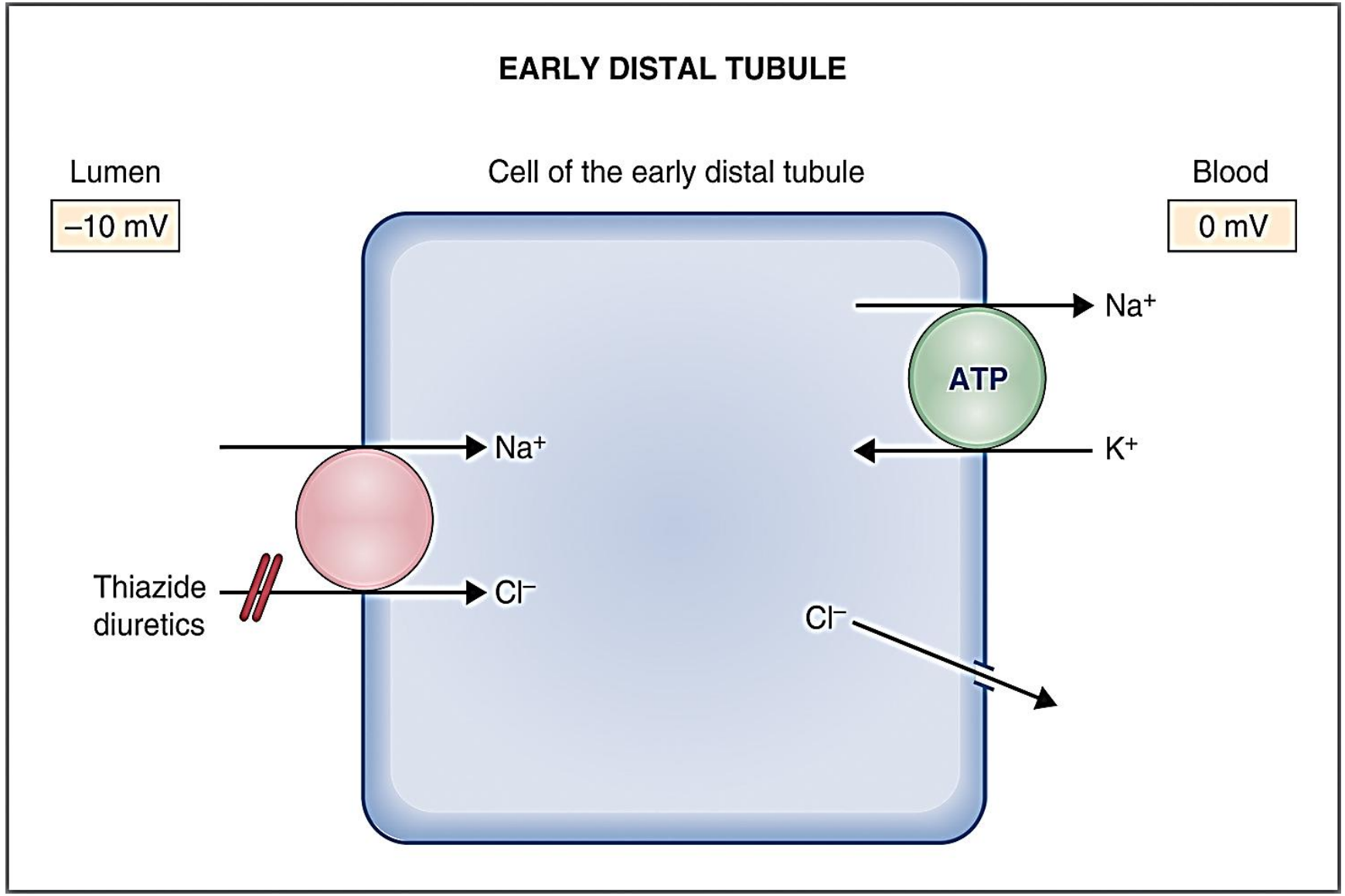
Early DCT: NCC Cotransporter

- **The Cortical Diluting Segment:** Reabsorbs 5% of filtered Na^+ and Cl^-; remains impermeable to water.

- **Cellular Mechanism:**
  - Apical electroneutral NCC cotransporter imports 1 Na^+ and 1 Cl^- down the chemical gradient.
  - Na^+ exits via basolateral Na^+/K^+-ATPase; Cl^- exits via basolateral chloride channels.
  - Luminal fluid osmolarity drops further (~80 mOsm/L).

- **Thiazide Diuretics:**
  - Compete for the Cl^--binding site on the NCC cotransporter.
  - Produce moderate diuresis (5% filtered Na^+ capacity).

- **Unique Feature: Dissociation of Na^+ and Ca^{2+} Handling**
  - Unlike the PCT and TAL, distal Na^+ and Ca^{2+} handling are **uncoupled**.
  - Thiazides **increase Ca^{2+} reabsorption** (useful in treating idiopathic hypercalciuria and calcium oxalate nephrolithiasis).


EARLY DISTAL TUBULE
Lumen
–10 mV
Cell of the early distal tubule
Blood
0 mV
Na+
ATP
K+
Na+
Thiazide
diuretics
Cl–
Cl–

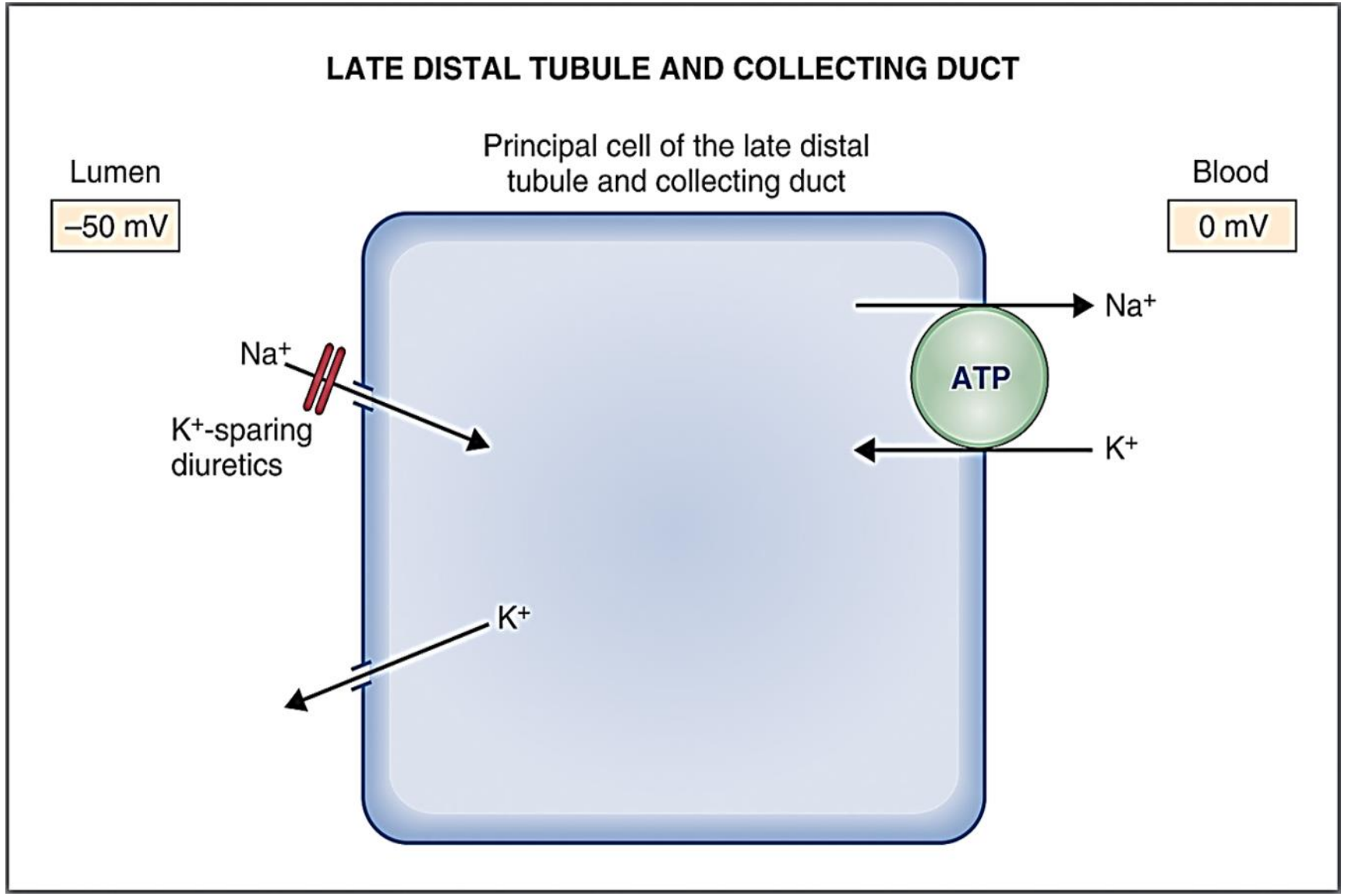
Late DCT & Collecting Duct: Principal Cells

- **The Master Regulator of Balance:** Principal cells reabsorb 3% of filtered Na^+ via uncoupled apical diffusion channels.

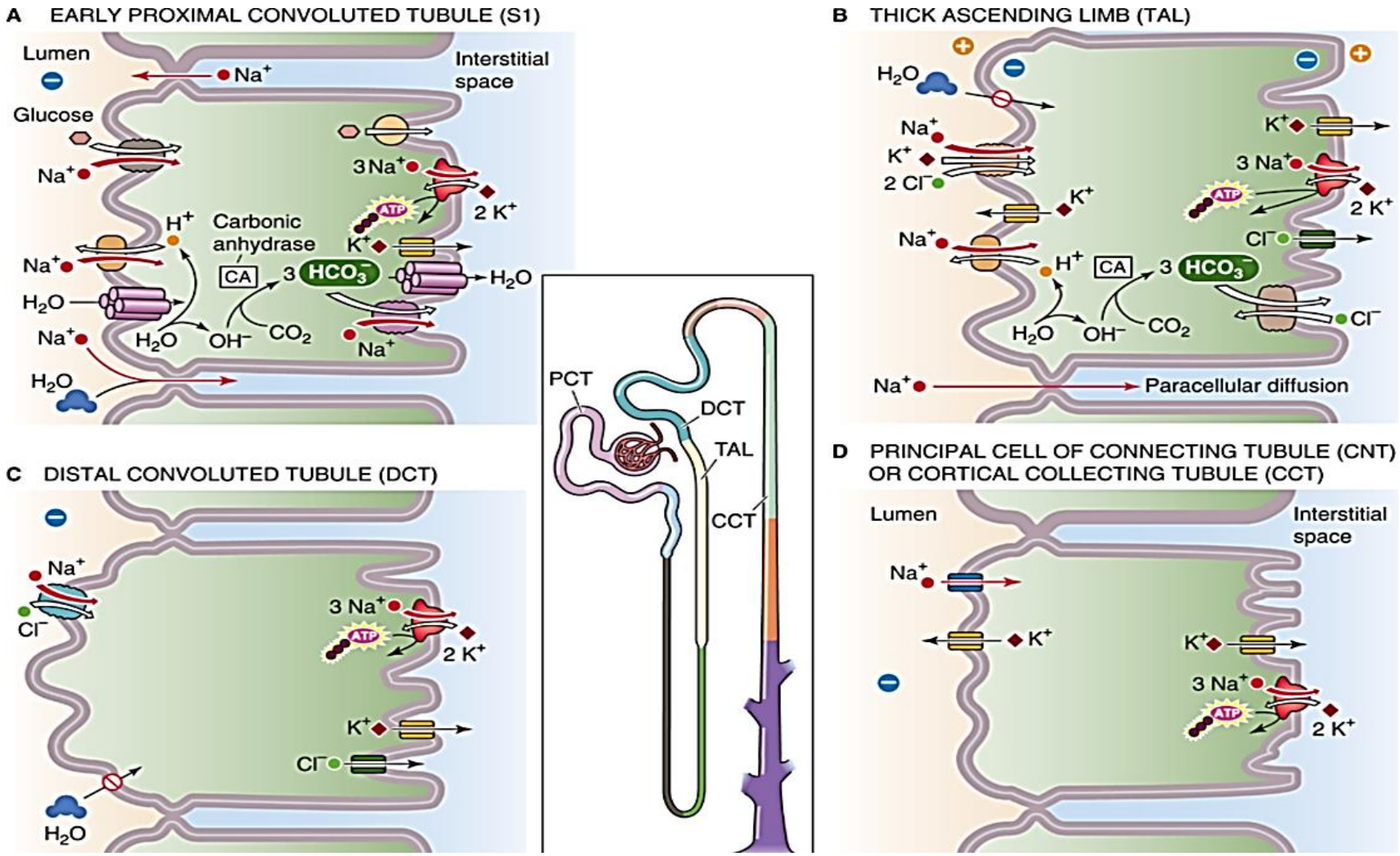
- **Cellular Architecture:**
  - Na^+ diffuses down its electrochemical gradient into the cell through **ENaC channels**.
  - Basolateral Na^+/K^+-ATPase extrudes Na^+ into blood while pumping K^+ into the cell.

- **The -50 mV Lumen-Negative Potential:**
  - Rapid electrogenic entry of Na^+ leaves a negative charge in the lumen (-50 mV).
  - This lumen electronegativity creates the electrical gradient that **pulls K^+ out of principal cells (ROMK)** and **pulls H^+ out of intercalated cells**.

- **Potassium-Sparing Diuretics:**
  - *ENaC Blockers (Amiloride, Triamterene):* Direct inhibition of apical Na^+ channels.
  - *Aldosterone Antagonists (Spironolactone, Eplerenone):* Competitive antagonists of cytoplasmic mineralocorticoid receptors.
  - **Physiological Result:** Mild natriuresis combined with **hyperkalemia** (loss of lumen-negative potential prevents K^+ secretion).


LATE DISTAL TUBULE AND COLLECTING DUCT
Principal cell of the late distal tubule and collecting duct
Lumen
–50 mV
Blood
0 mV
Na+
K+-sparing diuretics
ATP
Na+
K+
K+

Summary of Na^+ Reabsorption



Master Summary: Segmental Na^+ Transporters & Diuretics

Nephron Segment	Filtered Na^+ Reabsorbed	Apical Transporter	Transepithelial Potential	Diuretic Class	Molecular Drug Target	Key Electrolyte Side Effects
Early PCT	~67% (Total PCT)	SGLT2, Na^+-amino acid, NHE3	-4 mV (lumen negative)	Carbonic Anhydrase Inhibitors	Acetazolamide (inhibits CA)	Mild natriuresis, HCO_3^- wasting, metabolic acidosis
Late PCT	(Part of 67% above)	NHE3 + Cl^-/formate$^-$; paracellular	+4 mV (lumen positive)	Osmotic Diuretics	Mannitol (luminal osmotic draw)	Volume contraction, hypernatremia (early) / hyponatremia
Thick Ascending Limb	~25%	NKCC2 (1 Na^+ : 1 K^+ : 2 Cl^-)	+7 mV (lumen positive)	Loop Diuretics	Furosemide, Torsemide (Cl^- site on NKCC2)	Hypokalemia, hypocalcemia, hypomagnesemia, alkalosis
Early DCT	~5%	NCC (Na^+-Cl^- cotransporter)	-10 mV (lumen negative)	Thiazide Diuretics	HCTZ, Chlorthalidone (Cl^- site on NCC)	Hypokalemic metabolic alkalosis, hypercalcemia
Late DCT & CD	~3%	ENaC (epithelial Na^+ channel)	-50 mV (lumen negative)	K^+-Sparing Diuretics	Amiloride (ENaC); Spironolactone (MR)	Hyperkalemia, metabolic acidosis

Interim Summary 3

- **Thick Ascending Limb (TAL):** Reabsorbs 25% of filtered Na^+ via apical electroneutral **NKCC2** (1 Na^+ : 1 K^+ : 2 Cl^-). Apical K^+ back-leak through ROMK creates a **+7 mV lumen-positive potential** that drives paracellular **Ca^{2+} and Mg^{2+} reabsorption**.
  - *Loop Diuretics (Furosemide):* Inhibit the Cl^- site on NKCC2 → collapse the +7 mV potential → cause profound wasting of Na^+, Cl^-, K^+, Ca^{2+}, and Mg^{2+}.

- **Early Distal Tubule:** Reabsorbs 5% of filtered Na^+ and Cl^- via apical electroneutral **NCC**. Impermeable to water (cortical diluting segment).
  - *Thiazide Diuretics (HCTZ):* Inhibit NCC → promote natriuresis while **increasing Ca^{2+} reabsorption** (uncoupled distal Ca^{2+} handling).

- **Late Distal Tubule & Collecting Duct (Principal Cells):** Reabsorbs the final 3% of Na^+ through apical **ENaC channels**, generating a **-50 mV lumen-negative potential** that pulls K^+ out via ROMK and H^+ via intercalated cells.
  - *K^+-Sparing Diuretics:* Amiloride/Triamterene block ENaC; Spironolactone/Eplerenone block mineralocorticoid receptors → promote mild natriuresis while causing **hyperkalemia**.

Effective Arterial Blood Volume (EABV) & Sensors

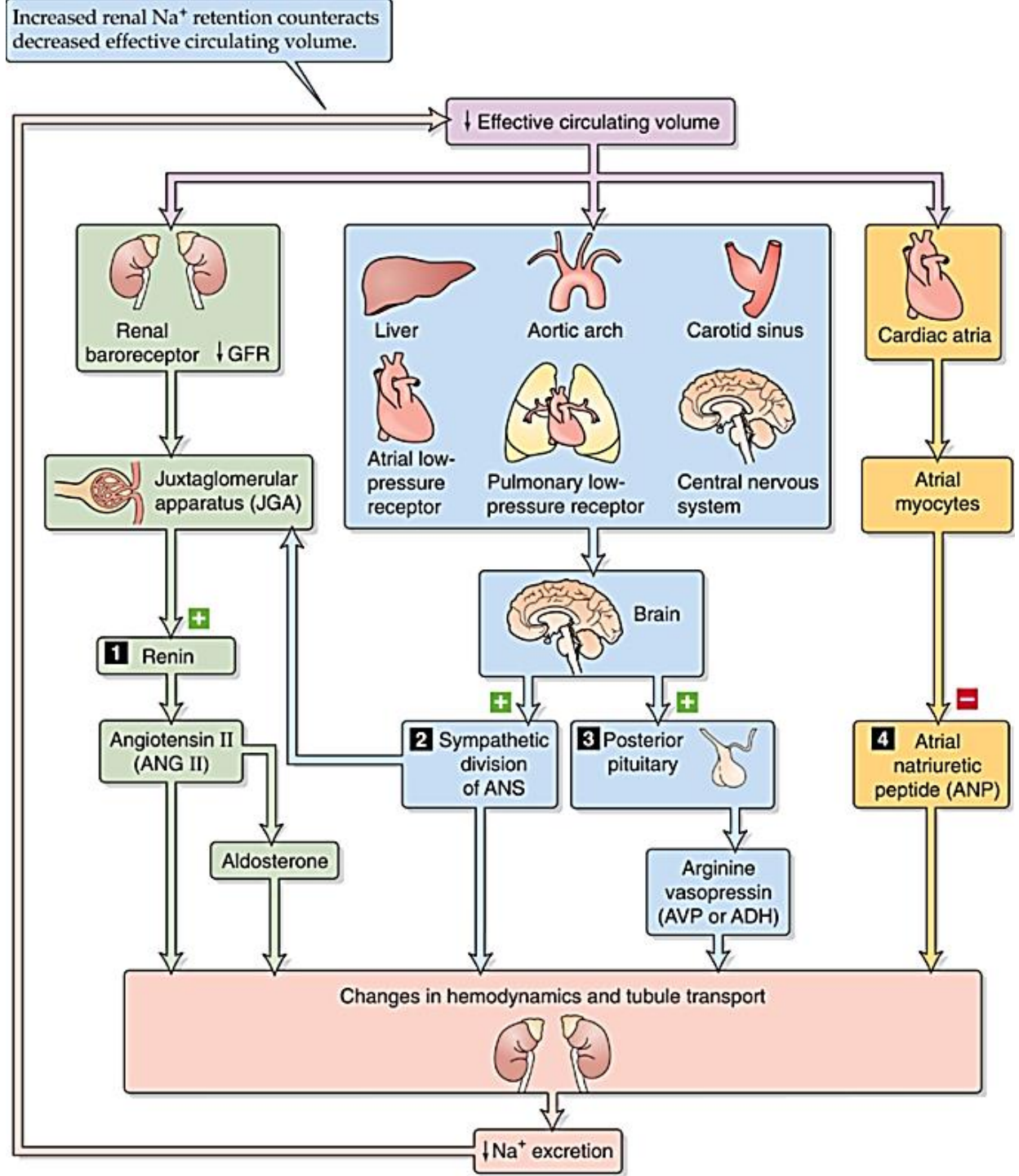
- **EABV Defined:** The effective fullness/perfusion of the arterial circulation as sensed by pressure/volume receptors.

- **ECF Volume vs. EABV Discordance (Clinical Pearl):**
  - In healthy states, ECF volume and EABV correlate directly (↑ ECF → ↑ EABV).
  - In **Heart Failure, Cirrhosis, and Nephrotic Syndrome**, total ECF volume is expanded (edema/ascites), but pooling or low cardiac output causes **decreased EABV**.
  - The kidneys sense low EABV as "underfilling" and inappropriately retain Na^+ and water, worsening the edema.

Effective Arterial Blood Volume (EABV) & Sensors

The Four Regulatory Effector Arms:

- *Renal Sympathetic Nerves:* Activated by low EABV via arterial baroreceptors.
- *RAAS Cascade:* Activated by renal hypoperfusion.
- *Atrial Natriuretic Peptide (ANP):* Suppressed by low EABV; activated by atrial distention.
- *Peritubular Capillary Starling Forces:* Regulated by systemic hemodynamics.


Increased renal Na+ retention counteracts decreased effective circulating volume.
↓ Effective circulating volume
Renal baroreceptor ↓GFR
Liver
Aortic arch
Carotid sinus
Cardiac atria
Atrial low-pressure receptor
Pulmonary low-pressure receptor
Central nervous system
Juxtaglomerular apparatus (JGA)
Atrial myocytes
Brain
1 Renin
2 Sympathetic division of ANS
3 Posterior pituitary
4 Atrial natriuretic peptide (ANP)
Angiotensin II (ANG II)
Aldosterone
Arginine vasopressin (AVP or ADH)
Changes in hemodynamics and tubule transport
↓Na+ excretion

The Renin-Angiotensin-Aldosterone System (RAAS)

- **Enzymatic Cascade:**
  - Renin (from JG cells) cleaves hepatic Angiotensinogen (453 aa) → Angiotensin I (10 aa).
  - Angiotensin-Converting Enzyme (ACE, pulmonary vascular endothelium) cleaves Ang I → Angiotensin II (8 aa).

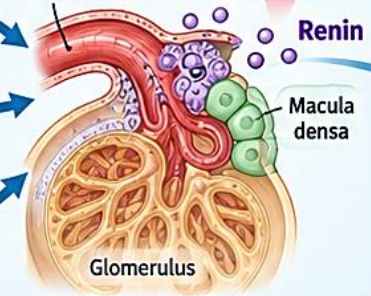
- **Direct Renal Actions of Angiotensin II:**
  - *Proximal Tubule:* Stimulates apical **NHE3 antiporters**, driving isosmotic Na^+, HCO_3^-, and water reabsorption.
  - *Efferent Arteriolar Constriction:* Preferentially constricts efferent arterioles → ↑ P_{GC} to sustain GFR while ↓ RPF, markedly driving up Filtration Fraction (↑ FF).
  - Elevated FF concentrates peritubular proteins (↑ π_c), further augmenting proximal fluid reabsorption.

- **Systemic Actions:** Systemic arteriolar vasoconstriction and direct stimulation of the adrenal cortex to release Aldosterone

Renin–Angiotensin–Aldosterone System (RAAS)

Triggers for Renin Release (JG Cells)

- Decreased renal perfusion pressure (↓ afferent arteriolar stretch)
- Decreased NaCl delivery to the macula densa
- Increased sympathetic stimulation (β_1-adrenergic)



Liver

Angiotensinogen (plaswa precursor)

Renin (cleaves angiotensinogen)

Angiotensin I (10 amino acids)

ACE (mainly lung & vascular endothelium)

Lungs (ACE)

Angiotensin II (8 amino acids)

Major Actions of Angiotensin II

Arterioles (systemic circulation)

- Vasoconstriction
- ↑ Total peripheral resistance (TPR)
- ↑ Blood pressure

Adrenal cortex (zona glomerulosa)

- ↑ Aldosterone secretion

Proximal tubule (kidney)

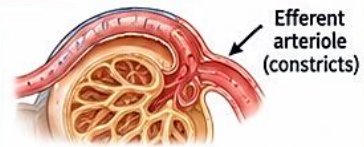
- ↑ Na^+ reabsorption (e.g., ↑ Na^+/H^+ exchange)
- ↑ Na^+ and water reabsorption

Posterior pituitary / hypothalamus

- ↑ ADH release
- ↑ Thirst

↑ Water reabsorption in collecting ducts (AQP2 insertion)

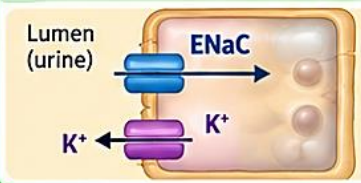
Kidney efferent arteriole



- Preferential constriction of efferent arteriole
- Helps maintain glomerular capillary pressure and GFR during low perfusion

Aldosterone Acts on Principal Cells

(late DCT / cortical collecting duct)



- ↑ ENaC-mediated Na^+ reabsorption
- ↑ Water retention (indirectly)
- ↑ K^+ secretion

→ ↑ ECF volume and ↑ Blood pressure

Net RAAS Effects

- ↑ Na^+ retention
- ↑ Water retention
- ↑ ECF volume
- ↑ Arterial pressure

Restoration of renal perfusion pressure and EABV suppresses renin release (negative feedback)

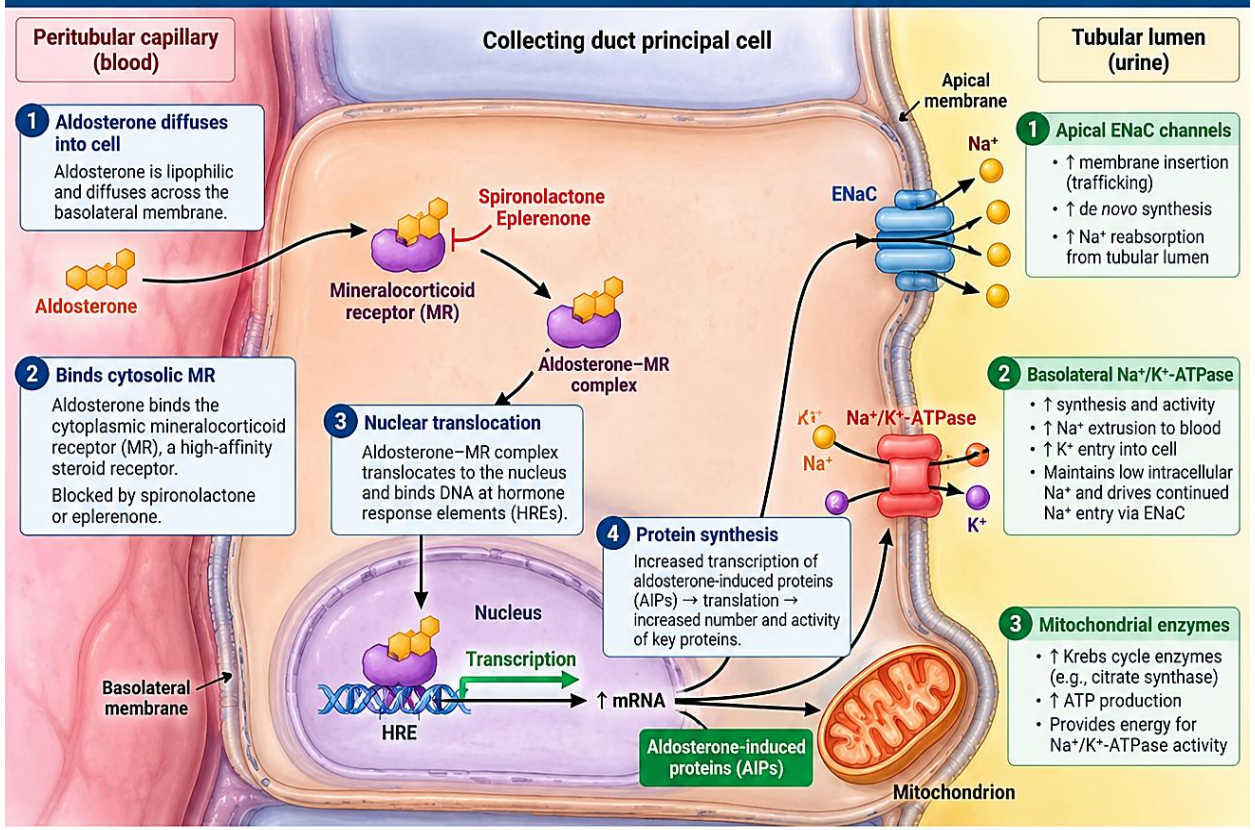
Genomic Action of Aldosterone in Principal Cells

- **Mechanism of Action:**
  - Aldosterone is a lipophilic steroid synthesized in the adrenal zona glomerulosa.
  - Diffuses across the basolateral membrane and binds the cytoplasmic mineralocorticoid receptor (MR).
  - Complex translocates to the nucleus to initiate transcription of specific mRNAs.

- **Triad of Cellular Effects:**
  - *Upregulates Apical ENaC:* Increases luminal Na^+ permeability, accelerating Na^+ entry.
  - *Upregulates Basolateral Na^+/K^+-ATPase:* Increases primary active extrusion of Na^+ into blood and uptake of K^+.
  - *Increases Luminal K^+ Permeability:* Upregulates ROMK channels; the elevated lumen-negative potential (-50 mV) drives K^+ secretion.

Aldosterone Action in a Collecting Duct Principal Cell

Mechanism and Aldosterone-Induced Proteins (AIPs)



Key Features of the Mineralocorticoid Receptor (MR)

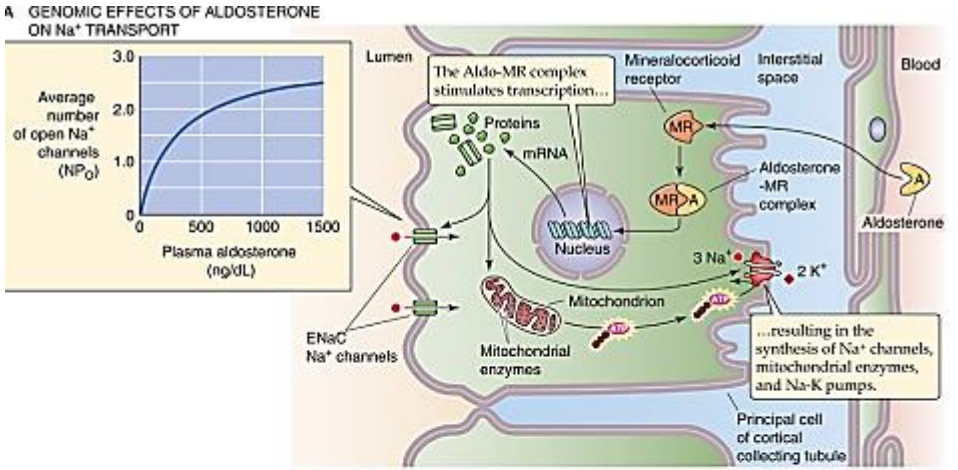
- Cytoplasmic, high-affinity steroid receptor
- Binds aldosterone (and cortisol, but 11β-HSD2 inactivates cortisol in principal cells)
- Blocked by spironolactone and eplerenone (competitive antagonists)
- Aldosterone–MR complex translocates to nucleus and regulates gene transcription

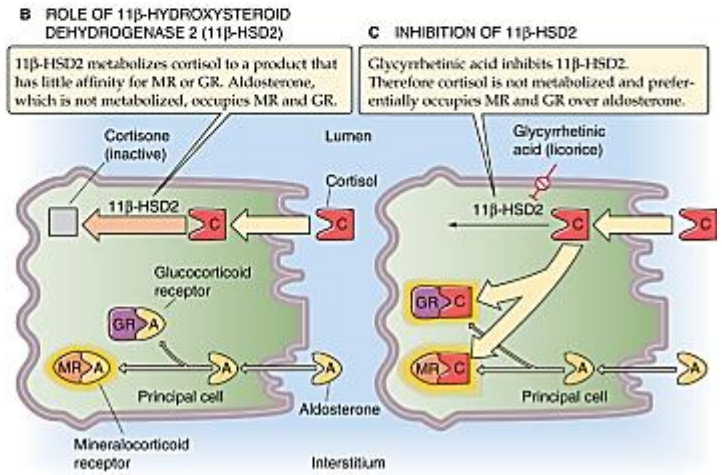
Aldosterone-Induced Proteins (AIPs)

1. **Apical ENaC channels**
   ↑ membrane insertion and de *novo* synthesis
2. **Basolateral Na⁺/K⁺-ATPase pumps**
   ↑ synthesis and activity
3. **Mitochondrial Krebs cycle enzymes**
   (e.g., citrate synthase)
   ↑ ATP production to fuel Na⁺/K⁺-ATPase

Net Effect

- ↑ Na⁺ reabsorption from tubular lumen
- ↑ K⁺ secretion into tubular lumen
- ↑ Water reabsorption (indirectly, as Na⁺ is reabsorbed)
- Helps maintain ECF volume and arterial pressure
- Excess aldosterone → Na⁺ retention, hypokalemia, metabolic alkalosis





Atrial Natriuretic Peptide (ANP) & BNP

- **The Endogenous Counter-Regulatory Hormone:** Synthesized and stored in atrial myocytes; secreted in response to hypervolemia and atrial stretch.
  - *Brain Natriuretic Peptide (BNP):* Released by ventricular myocytes under stretch; serves as a clinical biomarker for congestive heart failure.

- **Unique Hemodynamic Mechanism:**
  - **Dilates afferent arterioles and constricts efferent arterioles.**
  - Both actions act synergistically to **markedly increase glomerular capillary pressure (↑ P_{GC}) and GFR**.
  - Drives a higher filtered load of Na^+ into the nephron.
  - **ANP → afferent dilation + efferent constriction + ↑ K_f → ↑ GFR**

Atrial Natriuretic Peptide (ANP) & BNP

- **Tubular & Neurohumoral Actions:**
  - Directly inhibits Na^+ reabsorption in the medullary collecting duct.
  - Inhibits renin secretion from JG cells, breaking the RAAS cascade.
  - Directly inhibits aldosterone synthesis in the adrenal cortex.

- **Net Effect:** Marked natriuresis and diuresis, reducing ECF volume and arterial pressure back to baseline.

Cardiac–Renal Axis: ANP and Natriuresis

1 Increased ECF volume

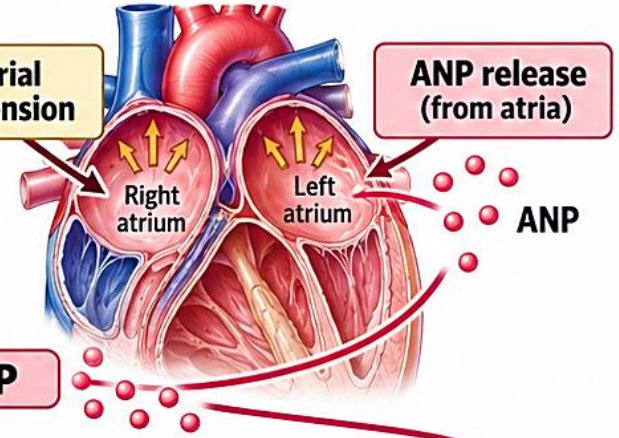
↑ ECF volume (extracellular fluid volume)

2 Increased venous return

↑ Venous return to heart

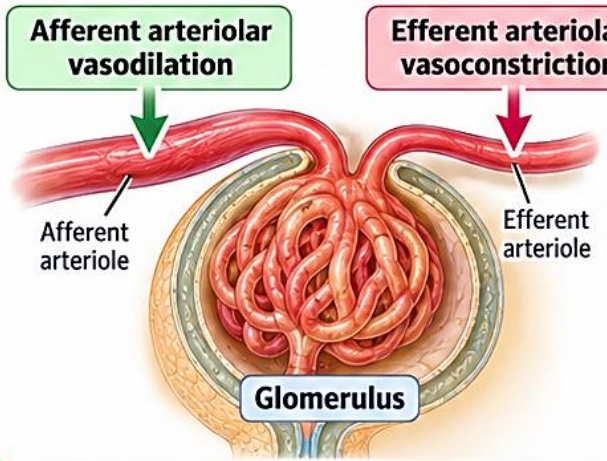
3 Mechanical stretch of cardiac atria

Increased venous return stretches the atrial walls (atrial distension).



A Renal hemodynamic action

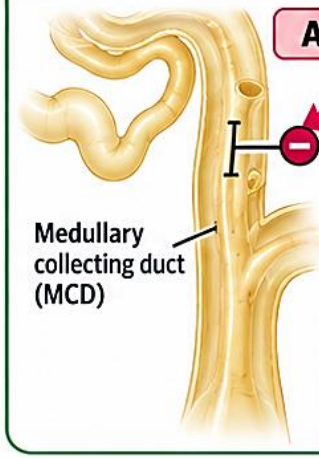
Altered arteriolar tone → ↑ GFR → ↑ filtered Na^+ load



↑ GFR
↓
↑ Filtered Na^+ load
↓
Promotes natriuresis

B Tubular action

Direct inhibition of Na^+ reabsorption in the medullary collecting duct



Direct inhibition of Na^+ reabsorption in the medullary collecting duct

↓ Na^+ reabsorption
↑ Na^+ excretion (natriuresis)
↑ Water excretion (diuresis)

Additional integrated effects of ANP (systemic)

Suppresses renin
↓ Aldosterone
Opposes volume expansion



Net effect: natriuresis + diuresis

↑ Na^+ and H_2O excretion (natriuresis + diuresis) → ↓ ECF volume (extracellular fluid volume) → ↓ venous return to heart → reduced atrial stretch (↓ ANP release)

Interim Summary 4

- **Effective Arterial Blood Volume (EABV):** The arterial fullness perfusing tissues. In heart failure, cirrhosis, and nephrotic syndrome, high total ECF volume coexists with **low EABV**, triggering paradoxical, maladaptive renal Na^+ retention.

- **The RAAS Axis:** Triggered by low afferent perfusion pressure, sympathetic β_1-stimulation, and decreased macula densa NaCl delivery.
  - *Angiotensin II:* Directly stimulates proximal NHE3 and constricts efferent arterioles (maintains GFR, elevates FF, and increases peritubular π_c).
  - *Aldosterone:* Genomic mechanism via cytoplasmic mineralocorticoid receptors (MR) that upregulates apical ENaC, basolateral Na^+/K^+-ATPase, and ROMK channels in principal cells.

- **Atrial Natriuretic Peptide (ANP):** Released by atrial stretch in response to volume expansion. Dilates afferent and constricts efferent arterioles (markedly increasing GFR and filtered Na^+ load), while directly inhibiting Na^+ reabsorption in the medullary collecting duct and shutting down renin and aldosterone.

Integrated Response to High Na^+ Intake

- **Sustaining Balance Under Dietary Variation:** Daily dietary sodium can range from 10 mEq/day to >300 mEq/day; the kidney adjusts excretion to match intake within 3 to 5 days.

Response to High Na^+ Diet (ECF Volume Expansion):

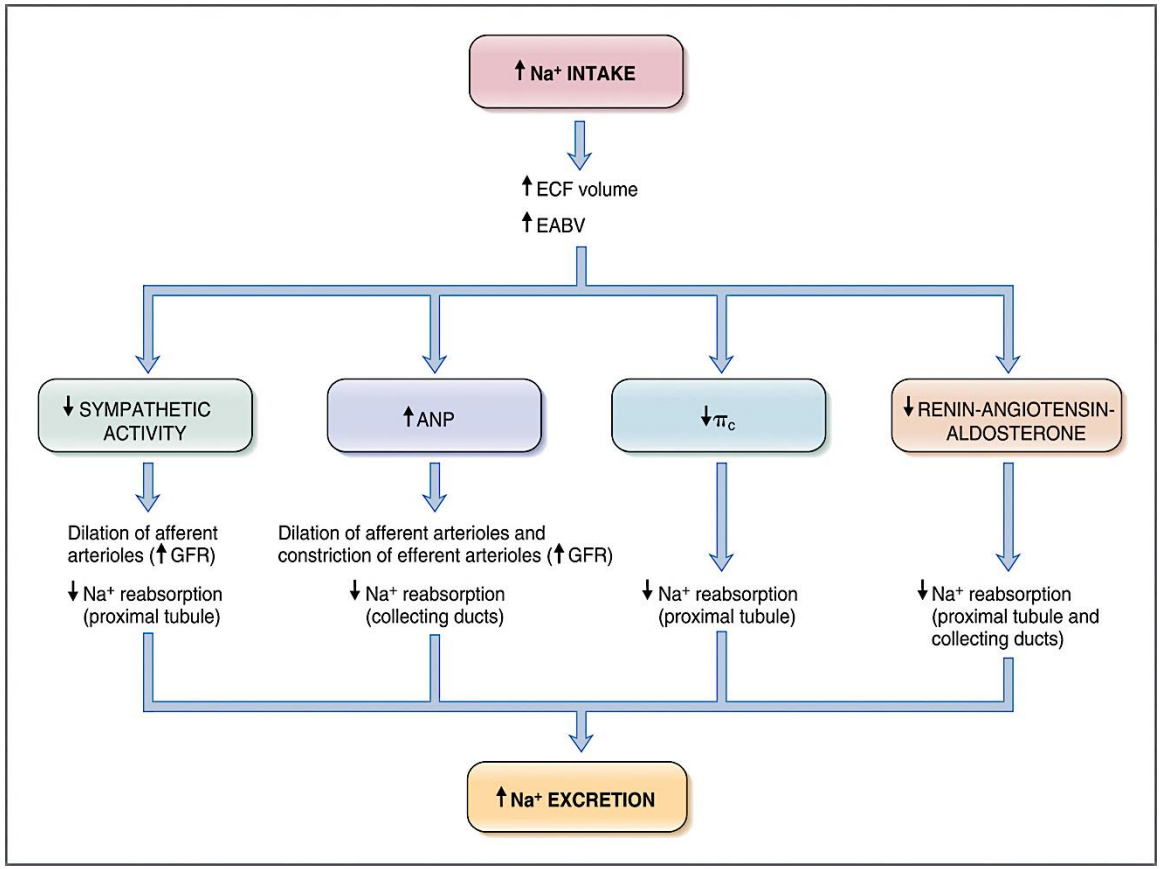
- *Sympathetic Tone Drops:* Afferent arteriole dilates (↑ GFR); proximal Na^+ reabsorption decreases.
- *ANP Increases:* Promotes afferent dilation/efferent constriction; blocks collecting duct Na^+ uptake.
- *Peritubular π_c Decreases:* Protein dilution suppresses proximal fluid uptake (GT balance shift).
- *RAAS Shuts Down:* Low Ang II and Aldosterone reduce proximal NHE3 and distal ENaC uptake.

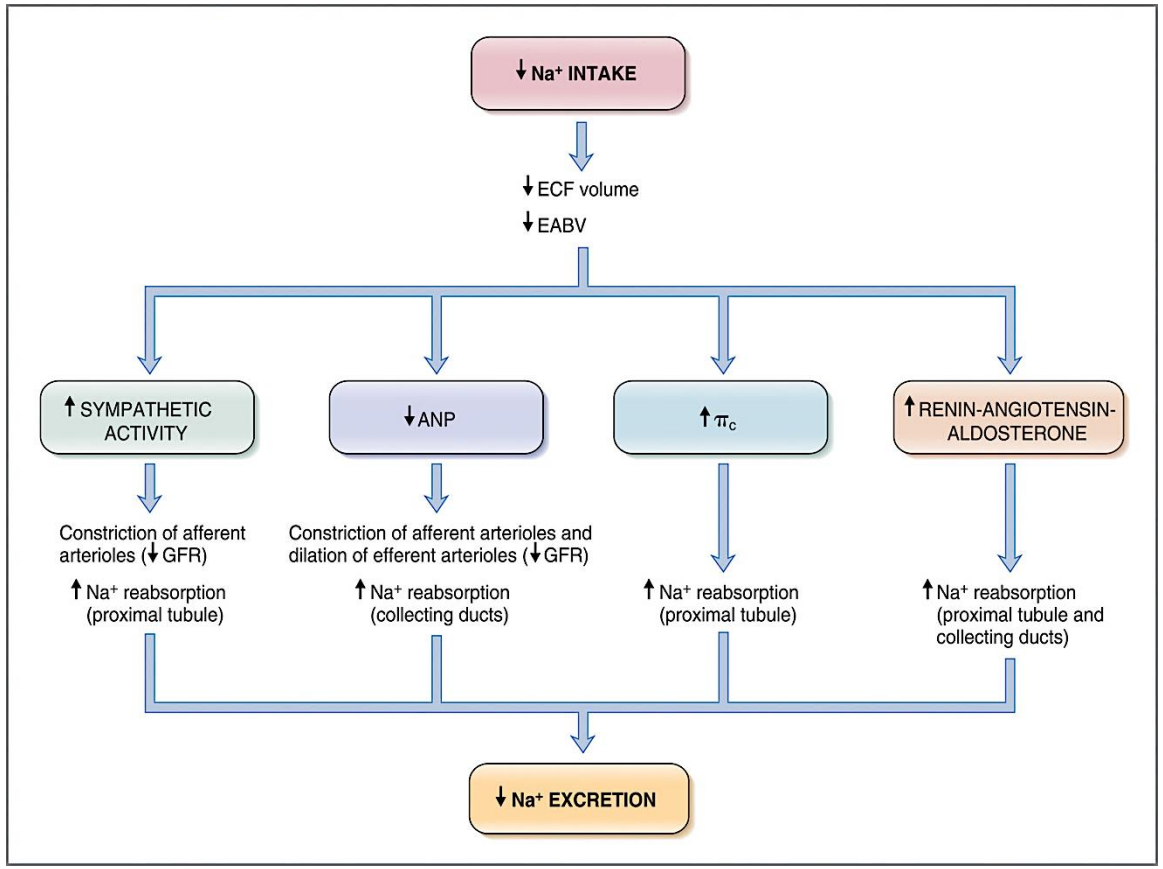
Integrated Response to Low Na^+ Intake

- **Sustaining Balance Under Dietary Variation:** Daily dietary sodium can range from 10 mEq/day to >300 mEq/day; the kidney adjusts excretion to match intake within 3 to 5 days.

Response to Low Na^+ Diet (ECF Volume Contraction):

- The exact reciprocal response activates: sympathetic vasoconstriction decreases RBF and usually GFR
- **Angiotensin II preferential efferent constriction** helps preserve glomerular pressure during low EABV
- RAAS drives maximal proximal and distal Na^+ retention, and urinary Na^+ excretion decreases significantly.


↑ Na+ INTAKE
↑ ECF volume
↑ EABV
↓ SYMPATHETIC ACTIVITY
↑ ANP
↓ π_c
↓ RENIN-ANGIOTENSIN-ALDOSTERONE
Dilation of afferent arterioles (↑ GFR)
Dilation of afferent arterioles and constriction of efferent arterioles (↑ GFR)
↓ Na+ reabsorption (proximal tubule)
↓ Na+ reabsorption (collecting ducts)
↓ Na+ reabsorption (proximal tubule)
↓ Na+ reabsorption (proximal tubule and collecting ducts)
↑ Na+ EXCRETION


↓ Na+ INTAKE
↓ ECF volume
↓ EABV
↑ SYMPATHETIC ACTIVITY
↓ ANP
↑ π_c
↑ RENIN-ANGIOTENSIN-ALDOSTERONE
Constriction of afferent arterioles (↓ GFR)
Constriction of afferent arterioles and dilation of efferent arterioles (↓ GFR)
↑ Na+ reabsorption (proximal tubule)
↑ Na+ reabsorption (collecting ducts)
↑ Na+ reabsorption (proximal tubule)
↑ Na+ reabsorption (proximal tubule and collecting ducts)
↓ Na+ EXCRETION

Primary Hyperaldosteronism & Mineralocorticoid Escape

- **Primary Hyperaldosteronism (Conn Syndrome):**
  - Caused by an aldosterone-secreting tumor of the adrenal zona glomerulosa.
  - **Hallmark Presentation:** Hypertension, hypokalemia, metabolic alkalosis, and suppressed plasma renin activity.

- **The Clinical Paradox:** Despite sky-high aldosterone levels, patients **do not develop generalized edema**. Why?

Primary Hyperaldosteronism & Mineralocorticoid Escape

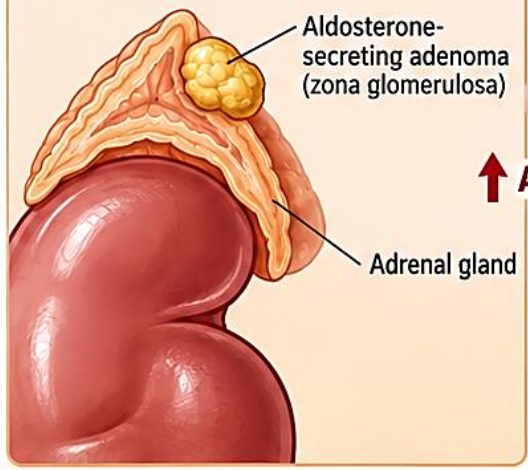
- **The Mechanism of "Aldosterone Escape":**
  - Aldosterone initially stimulates distal ENaC, causing positive Na^+ balance and ECF volume expansion.
  - ECF expansion elevates arterial pressure and dilutes peritubular oncotic pressure ($\downarrow \pi_c$).
  - Atrial stretch causes systemic release of ANP.
  - Decreased π_c and elevated ANP **inhibit proximal tubule Na^+ and water reabsorption**.
  - The reduced proximal Na^+ reabsorption increases distal Na^+ delivery until urinary Na^+ excretion again matches intake of sodium.

- **New Steady State:** Urinary Na^+ excretion matches intake; patients remain hypertensive, but "escape" edema formation.

Primary Hyperaldosteronism (Conn Syndrome): Mineralocorticoid Escape

Adrenal adenoma (Conn syndrome)

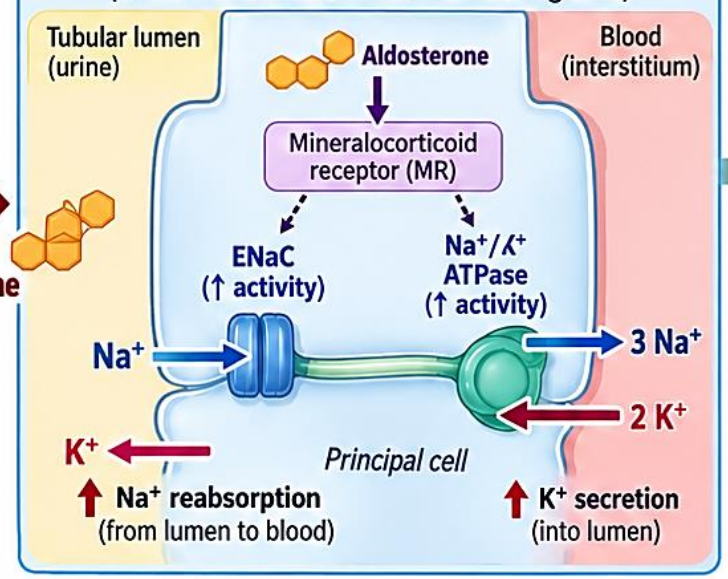
Autonomous, unregulated hypersecretion of aldosterone



↑ Aldosterone

Distal Principal Cell

(late distal tubule / cortical collecting duct)



Initial effects of aldosterone on distal nephron

- ↑ Na^+ and water reabsorption
- ↑ ECF volume
- ↑ Arterial blood pressure (hypertension)
- ↓ Plasma K^+ (hypokalemia)

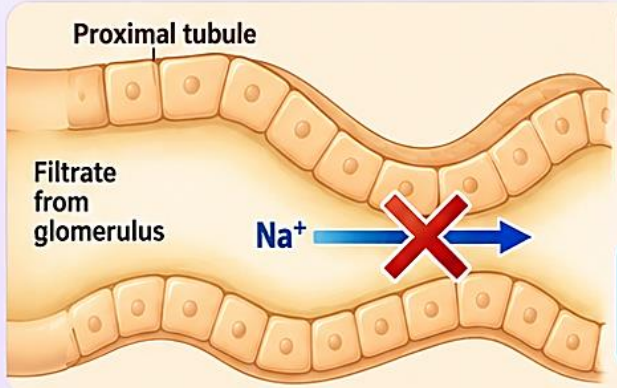
ECF volume expansion and hypertension



↑ Arterial blood pressure

Triggers mineralocorticoid escape
(reduces proximal Na^+ reabsorption)

Proximal Tubule Escape (Mineralocorticoid Escape)



ECF volume expansion:

- ↑ ANP
- ↓ peritubular reabsorptive forces (e.g., ↓ oncotic pressure, ↑ hydrostatic pressure)

↓ Proximal tubule Na^+ reabsorption (natriuresis)

Net Result in Conn Syndrome

- Urinary Na^+ excretion returns toward near-normal (mineralocorticoid escape)
- Continued K^+ secretion → **hypokalemia**
- **ECF volume expansion and hypertension**
- Persistent excess aldosterone (from autonomous adenoma)

Interim Summary 5

- **Dietary Sodium Adaptations:**
  - *High-Na$^+$Diet:* Suppresses sympathetic tone, elevates ANP, dilutes peritubular π_c, and shuts down RAAS → coordinated natriuresis restores normal volume.
  - *Low-Na$^+$Diet:* Activates sympathetic vasoconstriction, suppresses ANP, elevates peritubular π_c, and turns on RAAS → drops urinary Na^+ excretion to near zero.

- **Primary Hyperaldosteronism (Conn Syndrome):** Autonomous adrenal aldosterone secretion presents with hypertension, hypokalemia, and metabolic alkalosis, but **lacks generalized edema**.

- **Mechanism of Mineralocorticoid Escape:** Initial distal Na^+ retention causes ECF volume expansion. This secondary expansion dilutes peritubular π_c and triggers ANP release, which **decreases proximal tubule Na^+ reabsorption**. Proximal delivery overwhelms the distal collecting duct capacity, restoring urinary Na^+ excretion to match intake.

Segment	Percentage of Filtered NaCl Reabsorbed	Mechanism of Na^+ Entry Across Apical Membrane	Major Regulatory Hormones
Proximal tubule	67	Na^+-H^+ antiporter (NHE3), Na^+-glucose symporter (SGLT2), Na^+-symporter with amino acids, Cl^-/base antiporter (CFEX), paracellular	Angiotensin II, norepinephrine, epinephrine, dopamine
Loop of Henle	25	Na^+-K^+-$2Cl^-$ symporter (NKCC2), Na^+-H^+ antiporter (NHE3)	Aldosterone, angiotensin II
Distal tubule	~5	NaCl symporter (NCC)	Aldosterone, angiotensin II
Late distal tubule and collecting duct	~3	Na^+ channel (α, β, γ-ENaC)	Aldosterone, ANP, BNP, urodilatin, uroguanylin, guanylin, angiotensin II

Hormone*	Major Stimulus	Nephron Site of Action	Effect on Transport
Angiotensin II	↑ Renin	PT, TAL, DT/CD	↑ NaCl and H_2O reabsorption
Aldosterone	↑ Angiotensin II, ↑ $[K^+]_p$	TAL, DT/CD	↑ NaCl and H_2O reabsorption†
ANP, BNP, urodilatin	↑ ECFV	CD	↓ H_2O and NaCl reabsorption
Uroguanylin, guanylin	Oral ingestion of NaCl	PT, CD	↓ H_2O and NaCl reabsorption
Sympathetic nerves	↓ ECFV	PT, TAL, DT/CD	↑ NaCl and H_2O reabsorption
Dopamine	↑ ECFV	PT	↓ H_2O and NaCl reabsorption
AVP	↑ P_{osm}, ↓ ECFV	DT/CD	↑ H_2O reabsorption

Master Summary: Nephron Anatomy, Major Processes, and Key Functions

Re-anchoring sodium transport sites from PCT to collecting duct

Glomerulus / Bowman's capsule

Process: Filtration

- GFR ≈ 125 mL/min (≈ 180 L/day)
- Freely filtered water and small solutes (Na^+, K^+, Cl^-, HCO_3^-, glucose, amino acids, urea, etc.) enter filtrate

Proximal convoluted tubule (PCT)

Process: Bulk reabsorption (isosmotic reabsorption)

- Reabsorbs ≈ 65–70% of filtered Na^+ and H_2O
- **Key transporters:** NHE3, SGLT, Na^+/K^+ ATPase
- HCO_3^- reclamation (carbonic anhydrase)
- **Secretes** organic acids, bases, and drugs (e.g., PAH, penicillin)

Thin descending limb

Process: Water reabsorption

- Highly permeable to H_2O (AQP1)
- Relatively low NaCl permeability
- H_2O exits → tubular fluid becomes concentrated

Thick ascending limb (TAL)

Process: Diluting segment (builds medullary gradient)

- Reabsorbs ≈ 25% of filtered Na^+
- **Key transporters:** NKCC2 (Na^+-K^+-$2Cl^-$), ROMK (K^+ recycling)
- Impermeable to H_2O
- Reabsorbs Ca^{2+} and Mg^{2+} (paracellular)
- Establishes medullary gradient for urine concentration

Early distal convoluted tubule (DCT)

Process: Continued dilution

- Reabsorbs ≈ 5% of filtered Na^+
- **Key transporter:** NCC (Na^+-Cl^- cotransporter)
- Relatively impermeable to H_2O

Late DCT / Cortical collecting duct (principal cells)

Process: Fine-tuning of Na^+ and K^+

- **Key transporters:** ENaC (Na^+ entry), Na^+/K^+ ATPase
- **Aldosterone** ↑ Na^+ reabsorption and K^+ secretion

Collecting duct (intercalated cells)

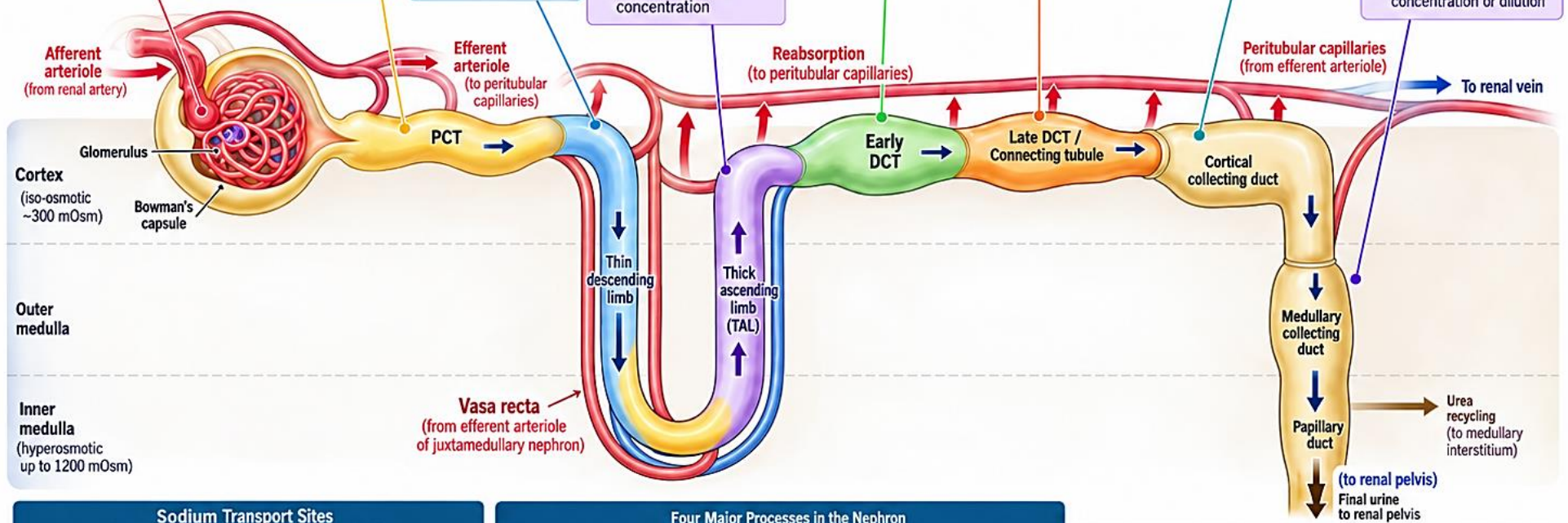
Process: Acid-base regulation

- Type A intercalated cells: secrete H^+ (H^+-ATPase, H^+/K^+ ATPase); reabsorb HCO_3^-
- Type B intercalated cells: secrete HCO_3^- (Pendrin); reabsorb H^+

Collecting duct overall (cortical → medullary → papillary duct)

Process: ADH-regulated water reabsorption and final urine concentration

- ADH ↑ H_2O permeability (via AQP2 in principal cells)
- Variable H_2O reabsorption
- Inner medullary collecting duct contributes to urea recycling (UT-A1, UT-A3)
- Determines final urine concentration or dilution



Sodium Transport Sites

(approximate % of filtered Na^+ reabsorbed)

Segment	% of filtered Na^+ reabsorbed
Proximal convoluted tubule (PCT)	65 – 70%
Thick ascending limb (TAL)	25%
Distal convoluted tubule (DCT)	5%
Collecting duct (late DCT + CD)	2 – 3%

Total reabsorbed: ~ 97 – 99%

Excreted in urine: normally < 1% of filtered Na^+

Four Major Processes in the Nephron

Process	Where it predominates	Key points
Filtration	Glomerulus	Forms filtrate (GFR ≈ 125 mL/min (≈ 180 L/day). Water and small solutes freely filtered.
Reabsorption	Throughout nephron (especially PCT)	Returns water and solutes to blood (Na^+, H_2O, HCO_3^-, glucose, amino acids, Ca^{2+}, etc.).
Secretion	PCT and distal nephron / collecting duct	Adds substances from peritubular capillaries to tubular fluid (H^+, K^+, organic acids/bases, drugs).
Excretion	Final urine leaving collecting duct (to renal pelvis)	Net elimination of water and solutes in urine.

Mass Balance Relationship

Excretion = Filtration − Reabsorption + Secretion

- **Filtration:** at glomerulus (GFR ≈ 125 mL/min or 180 L/day)
- **Reabsorption:** throughout nephron (especially PCT)
- **Secretion:** in PCT and distal nephron/collecting duct
- **Excretion:** as final urine from collecting duct

Master Summary

- **Sodium Balance Dictates Volume:** ECF volume and arterial pressure depend strongly on total body Na^+ content, defended by matching daily excretion to intake.

- **Segmental Coordination:**
  - *PCT (67%):* Isosmotic bulk retrieval governed by basolateral Na^+/K^+-ATPase and protected by glomerulotubular balance.
  - *TAL (25%):* Watertight NKCC2 cotransporter generating +7 mV potential; blocked by loop diuretics.
  - *Early DCT (5%):* Watertight NCC cotransporter; blocked by thiazides (spares Ca^{2+}).
  - *Late DCT/CD (3%):* ENaC channels regulated by aldosterone; blocked by K^+-sparing diuretics.

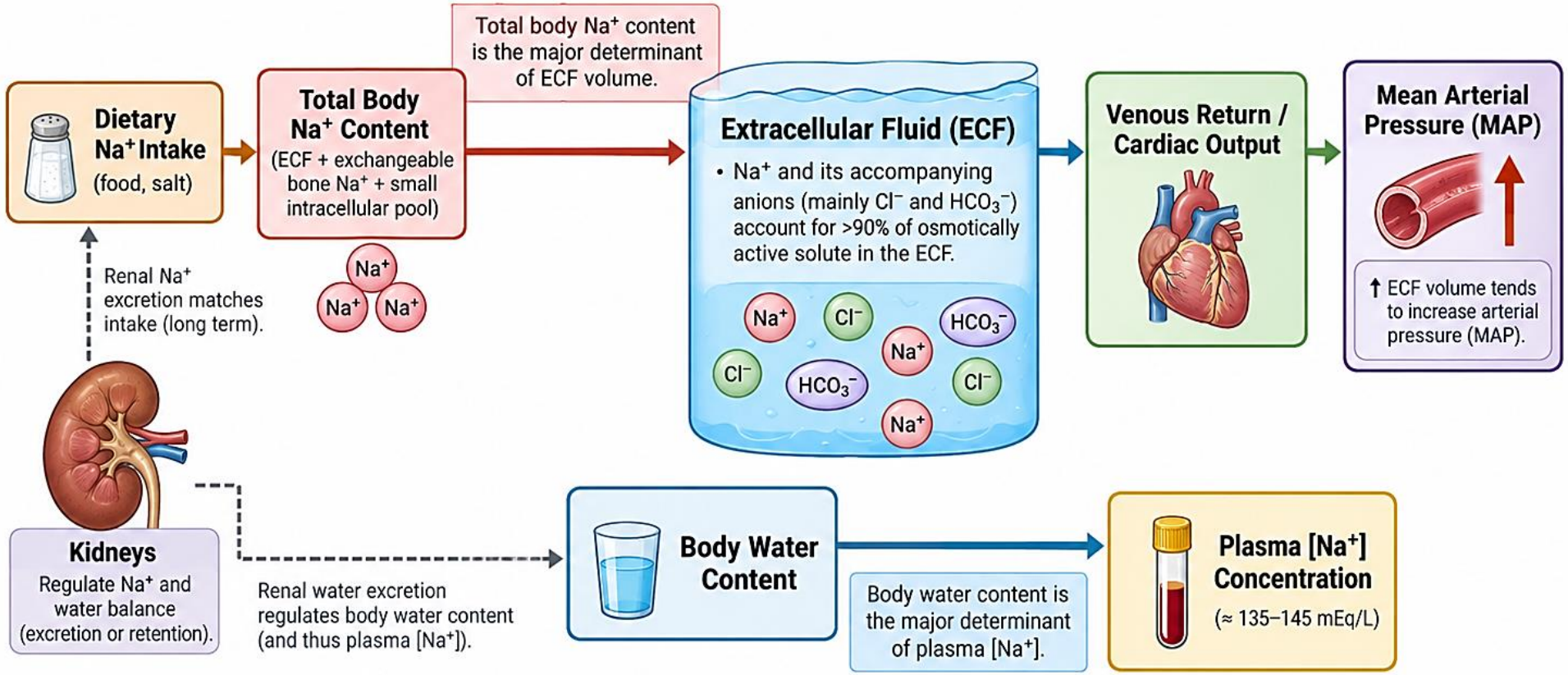
- **Endocrine Counter-Balance:** RAAS and sympathetic nerves conserve sodium to defend EABV; ANP promotes natriuresis during volume expansion.

- Total body Na^+ content is the major determinant of ECF volume; changes in ECF volume influence arterial pressure.

The End

Appendix Information

Reference materials, reading beyond the presentation, and anything else that needs to be included for completeness!



Volume vs Concentration

Total body Na⁺ content determines ECF volume.

Body water content determines plasma [Na⁺].

Example Changes

↑ Total body Na⁺ → ↑ ECF volume → ↑ MAP

(e.g., high salt intake or renal Na⁺ retention)

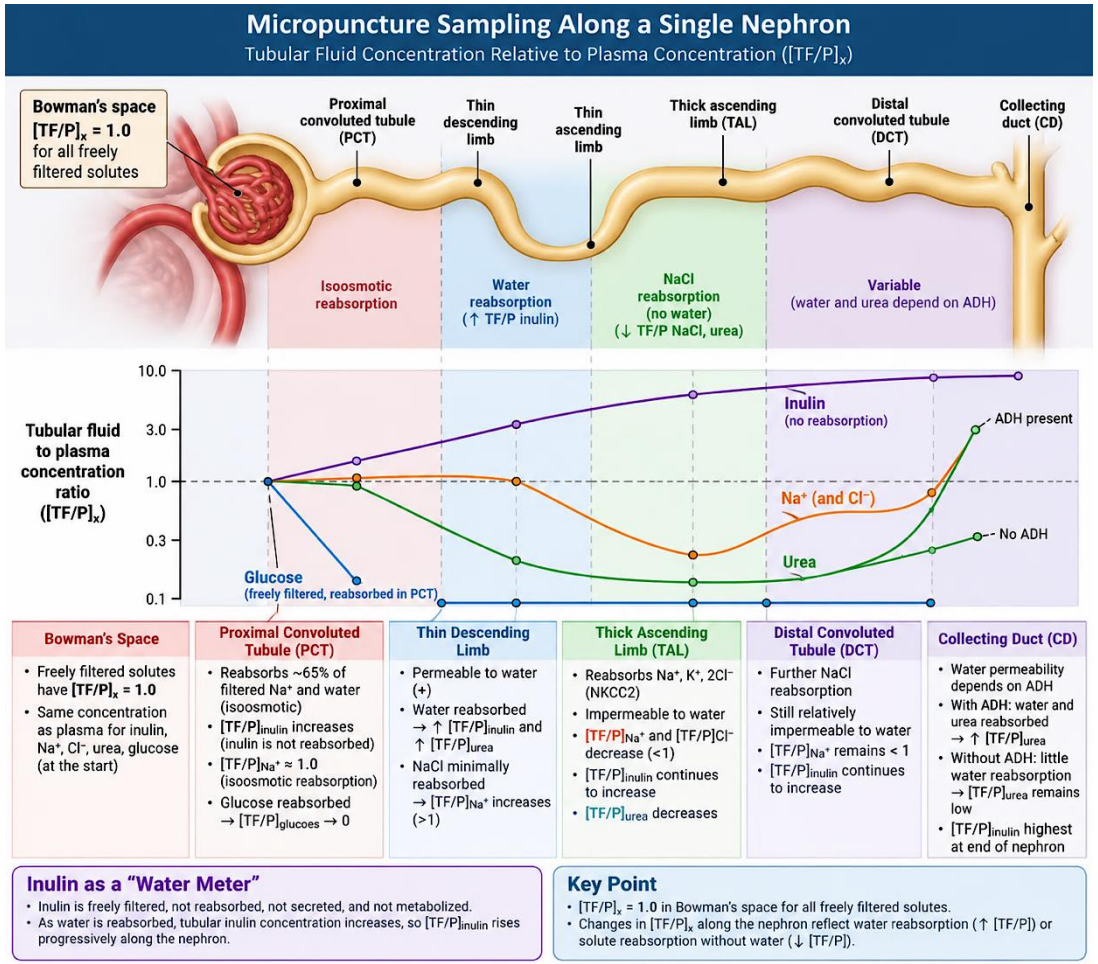
↑ Body water relative to Na⁺ → ↓ Plasma [Na⁺]

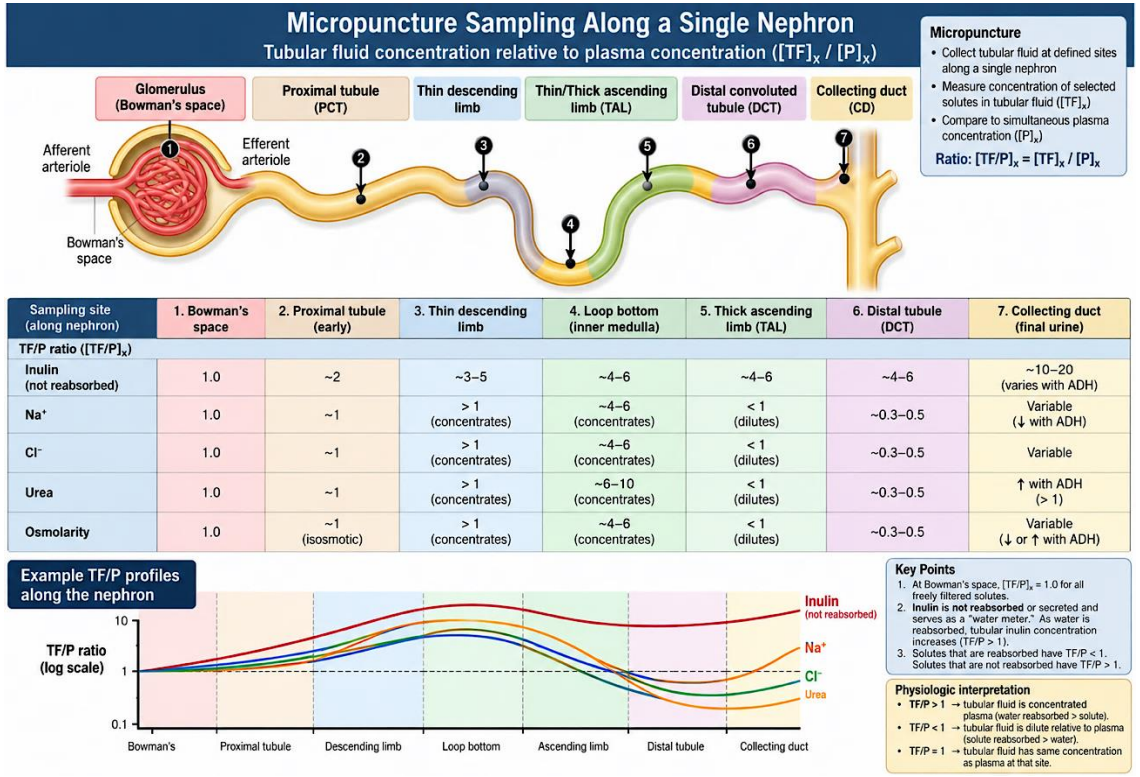
(e.g., excess water intake, SIADH)

Key Points

- Total body Na⁺ content is the major determinant of ECF volume.
- Body water content is the major determinant of plasma [Na⁺].
- Increased ECF volume increases venous return, cardiac output, and thus mean arterial pressure.
- Plasma [Na⁺] reflects the ratio of total body Na⁺ to body water, not the absolute total body Na⁺ content.

[TF/P] Ratio Concept





Sampling site (along nephron)	1. Bowman's space	2. Proximal tubule (early)	3. Thin descending limb	4. Loop bottom (inner medulla)	5. Thick ascending limb (TAL)	6. Distal tubule (DCT)	7. Collecting duct (final urine)
TF/P ratio ([TF/P]x)							
Inulin (not reabsorbed)	1.0	~2	~3–5	~4–6	~4–6	~4–6	~10–20 (varies with ADH)
Na^+	1.0	~1	> 1 (concentrates)	~4–6 (concentrates)	< 1 (dilutes)	~0.3–0.5	Variable (↓ with ADH)
Cl^-	1.0	~1	> 1 (concentrates)	~4–6 (concentrates)	< 1 (dilutes)	~0.3–0.5	Variable
Urea	1.0	~1	> 1 (concentrates)	~6–10 (concentrates)	< 1 (dilutes)	~0.3–0.5	↑ with ADH (> 1)
Osmolarity	1.0	~1 (isosmotic)	> 1 (concentrates)	~4–6 (concentrates)	< 1 (dilutes)	~0.3–0.5	Variable (↓ or ↑ with ADH)

Board-Style Clinical Vignette: Diuretic Resistance

- A 72-year-old male with a history of membranous nephropathy presents with severe bilateral lower extremity pitting edema, ascites, and a 5-kg weight gain over two weeks. Laboratory evaluation reveals:
  - Serum Albumin: 1.8 g/dL (Normal: 3.5–5.0 g/dL)
  - Urinalysis: 4+ Protein (nephrotic-range proteinuria)
  - Blood Pressure: 155/95 mmHg

- **Clinical Dilemma:** The patient is prescribed oral furosemide 40 mg daily, but after 3 days, there is no significant increase in urine output and his edema worsens (diuretic resistance).

Board-Style Clinical Vignette: Diuretic Resistance

- **Physiological Analysis:**
  - *Mechanism 1 (Drug Inactivation):* Furosemide is an organic acid that acts on the luminal Cl^- site of NKCC2. In nephrotic syndrome, massive proteinuria allows albumin to spill into the lumen, where it binds free furosemide and prevents it from reaching the transporter.
  - *Mechanism 2 (Distal Hypertrophy & RAAS):* Profound hypoalbuminemia lowers plasma oncotic pressure, shifting fluid into the interstitium and dropping EABV. Low EABV hyperactivates RAAS; high aldosterone stimulates distal ENaC, reclaiming any Na^+ that escapes the loop.

- **Management Strategy:**
  - Substantially increase the dose of loop diuretic to overwhelm intraluminal protein binding.
  - Add **Spironolactone** (aldosterone antagonist) to block distal ENaC compensation and break the cycle of sodium retention.

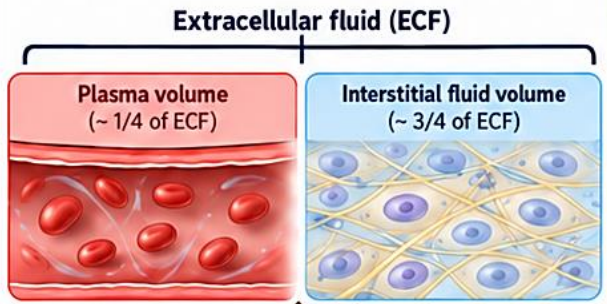
Extracellular Fluid (ECF) Volume vs Effective Arterial Blood Volume (EABV)

Total Extracellular Fluid (ECF) Volume

- ECF = plasma volume + interstitial fluid volume
- Total body sodium (Na^+) content determines ECF volume (water follows sodium).
- ECF provides the volume from which the circulation is filled, but it is not the same as effective arterial blood volume (EABV).

Total ECF volume

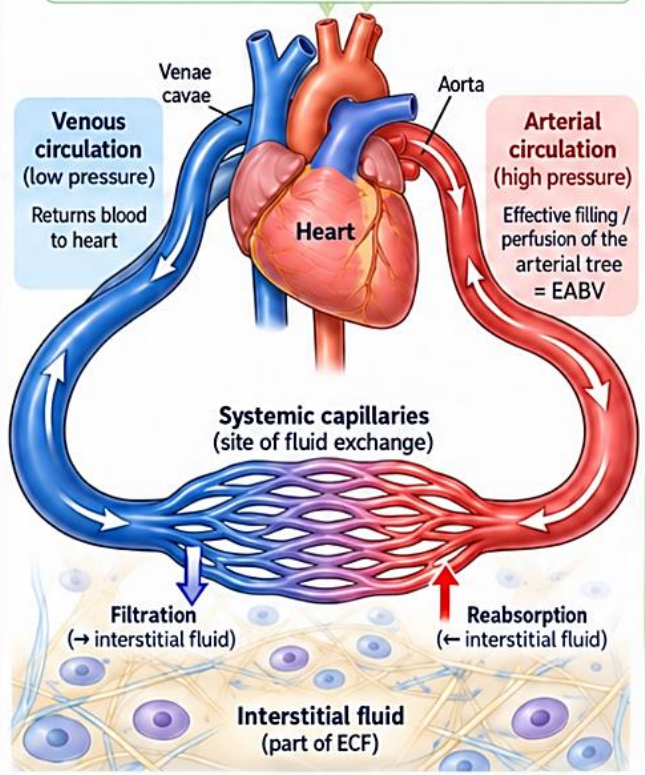
= Plasma volume + Interstitial fluid volume



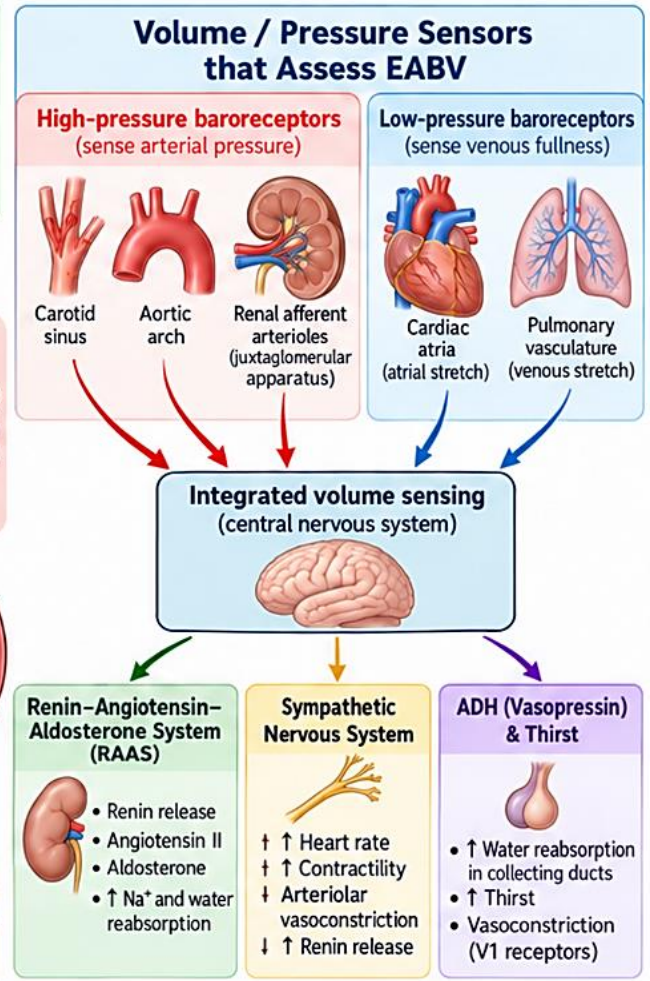
Total body Na^+ content determines ECF volume.

Effective Arterial Blood Volume (EABV)

- **EABV** = the portion of the circulation effectively perfusing the arterial tree and tissues.
- Reflects adequacy of arterial perfusion sensed by baroreceptors.
- Can be reduced even when total ECF volume is increased (e.g., heart failure, cirrhosis).

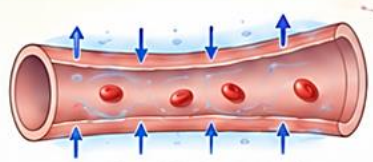


Volume / Pressure Sensors that Assess EABV



Conceptual Distinction: ECF Volume vs EABV

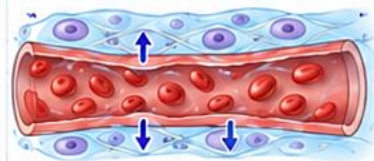
1. Normal or Volume Depletion



Reduced total body Na^+ and water
→ ↓ ECF and ↓ EABV

↓ ECF volume
→ ↓ plasma volume
→ ↓ **EABV**
(typically concordant)

2. Heart Failure or Cirrhosis



Increased total body Na^+ and water
→ ↑ ECF volume (edema)
but ↓ EABV (inadequate arterial perfusion)

↑ ECF volume
(both plasma and interstitial fluid)
but ↓ **EABV**
due to poor cardiac output or splanchnic vasodilation.

Key Takeaways

- Total ECF volume = plasma volume + interstitial fluid volume.
- ECF volume is determined by total body Na^+ content (water follows sodium).
- **EABV** = the portion of the circulation effectively perfusing the arterial tree and tissues.
- **EABV** reflects adequacy of arterial perfusion sensed by baroreceptors and can be reduced even when total ECF is increased (e.g., heart failure, cirrhosis).

EABV is not identical to total ECF volume.

Structure	Abbreviation	Meaning	Units and/or Normal Value
Whole Kidney	C	Clearance	mL/min
	[U]	Concentration in urine	mg/mL
	[P]	Concentration in plasma	mg/mL
	$\dot{V}$	Urine flow rate	mL/min
	GFR	Glomerular filtration rate	120 mL/min
	RPF	Renal plasma flow	660 mL/min
	RBF	Renal blood flow	1200 mL/min
Single Nephron	[TF]	Concentration in tubular fluid	mg/mL
	$[TF/P]_x$	Concentration of x in tubular fluid relative to concentration of x in plasma	None
	$[TF/P]_{inulin}$	Concentration of inulin in tubular fluid relative to concentration of inulin in plasma	None
	$[TF/P]_x/[TF/P]_{inulin}$	Fraction of the filtered load remaining in tubular fluid, or fractional excretion	None